

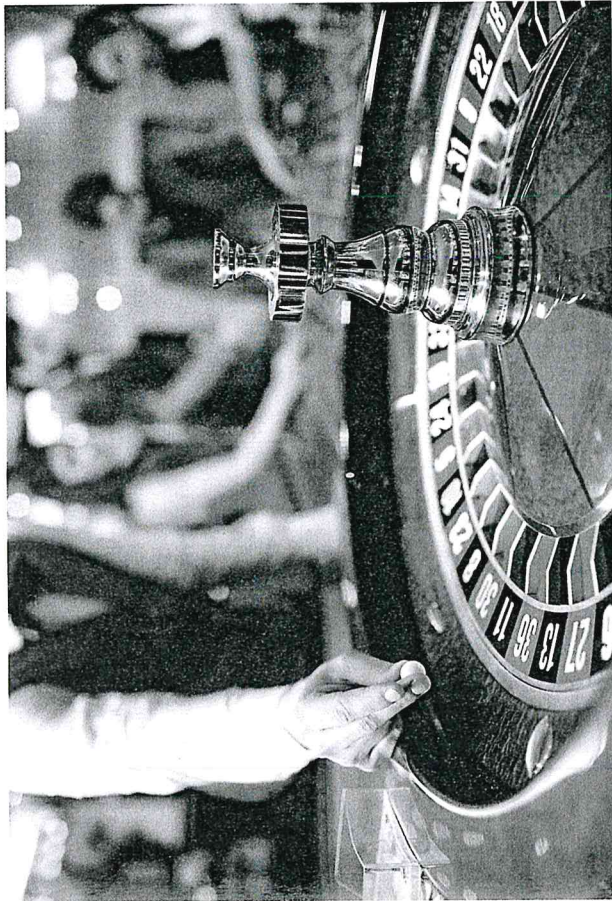
Introduction to Risk, Return, and the Opportunity Cost of Capital

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

- 11-1 Estimate the opportunity cost of capital for an "average-risk" project.
- 11-2 Calculate returns and standard deviation of returns for individual common stocks or for a stock portfolio.
- 11-3 Understand why diversification reduces risk.
- 11-4 Distinguish between specific risk, which can be diversified away, and market risk, which cannot.

RELATED WEBSITES FOR THIS CHAPTER CAN BE FOUND IN CONNECT.



Investing in risky assets is not the same as gambling. After reading this chapter, you should be able to explain the difference and understand the rewards for bearing risk in the stock market. *Studio Romantic/Shutterstock*

In earlier chapters, we skirted the issue of project risk; now it is time to confront it head-on. We can no longer be satisfied with vague statements like, "The opportunity cost of capital depends on the risk of the project." We need to know how to measure risk, and we need to understand how risk affects the cost of capital. These are the topics of the next two chapters.

Think for a moment what the cost of capital for a project means. It is the rate of return that shareholders could expect to earn if they invested in equally risky securities. So one way to estimate the cost of capital is to find securities that have the same risk as the project and then estimate the expected rate of return on these securities.

We start our analysis by looking at the rates of return earned in the past from different investments, concentrating on the extra return that investors have received for investing in risky rather than safe securities. We then show how to measure the risk of a portfolio, and we look again at past history for a guide to the risk of stock market investments.

Finally, we explore the concept of diversification. Most investors do not put all their eggs into one basket—they diversify. Thus, investors are not concerned with the risk of each security in isolation; instead, they are concerned with how much it contributes to the risk of a diversified portfolio. Therefore, we need to distinguish between the risk that can be eliminated by diversification and the risk that cannot be eliminated.

11.1 Rates of Return: A Review

When investors buy a stock or a bond, their return comes in two forms: (1) a dividend or interest payment and (2) a capital gain or a capital loss. For example, you would have been very happy to have invested in Microsoft stock in 2020. Its stock price at the beginning of the year was \$158.78 a share. By the end of the year the value of that investment had appreciated to \$222.42, giving a capital gain of $\$222.42 - \$158.78 = \$63.64$. In addition, in 2020 Microsoft paid a dividend of \$2.09 a share.

The *percentage return* on your investment was, therefore,

$$\begin{aligned}\text{Percentage return} &= \frac{\text{capital gain} + \text{dividend}}{\text{initial share price}} \\ &= \frac{\$63.64 + \$2.09}{\$158.78} = .414 \text{ or } 41.4\% \quad (11.1)\end{aligned}$$

The percentage return can also be expressed as the sum of the *dividend yield* and *percentage capital gain*. The dividend yield is the dividend expressed as a percentage of the stock price at the beginning of the year:

$$\begin{aligned}\text{Dividend yield} &= \frac{\text{dividend}}{\text{initial share price}} \\ &= \frac{\$2.09}{\$158.78} = .0132 \text{ or } 1.32\%\end{aligned}$$

Similarly, the percentage capital gain is

$$\begin{aligned}\text{Percentage capital gain} &= \frac{\text{capital gain}}{\text{initial share price}} \\ &= \frac{\$63.64}{\$158.78} = .4008 \text{ or } 40.08\%\end{aligned}$$

Thus the total return is the sum of $1.32\% + 40.08\% = 41.4\%$, which is the same as we calculated above.

Remember that in Chapter 5, we made a distinction between the *nominal* rate of return and the *real* rate of return. The nominal return measures how much more money you will have at the end of the year if you invest today. The return that we just calculated is, therefore, a nominal return. The real rate of return tells you how much more you will be able to *buy* with your money at the end of the year. To convert from a nominal to a real rate of return, we use the following relationship:

$$1 + \text{real rate of return} = \frac{1 + \text{nominal rate of return}}{1 + \text{inflation rate}}$$

In 2020, inflation was 1.4%. So we calculate the real rate of return on Microsoft stock as follows:

$$1 + \text{real rate of return} = \frac{1.414}{1.014} = 1.394$$

Therefore, the real rate of return equals .394, or 39.4%. Inflation in 2020 was modest, so the real return was only slightly less than the nominal return.

11.1 Self-Test

Suppose you buy a bond for \$1,020 with a 15-year maturity paying an annual coupon of \$80. A year later, interest rates have dropped and the bond's price has increased to \$1,050. What are your nominal and real rates of return? Assume the inflation rate is 4%.

11.2 A Century of Capital Market History

When you invest in a stock, you don't know what return you will earn. But by looking at the history of security returns, you can get some idea of the return that investors expected from different types of securities and of the risks that they faced. Let us look, therefore, at the risks and returns that investors have experienced in the past.

Market Indexes

Investors can choose from an enormous number of different securities. For example, about 2,800 United States and foreign companies currently list their stocks on the New York Stock Exchange. About 3,300 stocks are traded on the NASDAQ stock market.

Financial analysts can't track every stock, so they rely on **market indexes** to summarize the return on different classes of securities. The best-known stock market index in the United States is the **Dow Jones Industrial Average**, generally known as the *Dow*. The Dow tracks the performance of a portfolio that holds one share in each of 30 large firms. For example, suppose that the Dow starts the day at a value of 31,000 and then rises by 310 points to a new value of 31,310. An investor with a \$31,000 investment in the Dow portfolio would make a capital gain of $\$310/\$31,000 = .01$, or 1%.

The Dow Jones Industrial Average was first computed in 1896. Most people are used to it and expect to hear it on the 6 o'clock news. However, it is far from the best measure of the performance of the stock market. First, with only 30 large stocks, it is not representative of the performance of stocks generally. Second, investors don't usually hold an equal number of shares in each company. For example, in 2020 there were 5.6 billion shares in Pfizer and .56 billion in Boeing. So, on average, investors did *not* hold the same number of shares in the two firms. Instead, they held 10 times as many shares in Pfizer as in Boeing. It doesn't make sense, therefore, to look at an index that measures the performance of a portfolio with an equal number of shares in the two firms.

The **Standard & Poor's Composite Index**, better known as the *S&P 500*, includes the stocks of 500 major companies and is, therefore, a more comprehensive index than the Dow. Also, it measures the performance of a portfolio that holds shares in each firm in proportion to the number of shares that are held by investors. For example, the S&P 500 portfolio would hold 10 times as many shares in Pfizer as in Boeing. Thus, the S&P 500 shows the *average* performance of investors in the 500 firms.

Only a small proportion of the publicly traded companies are represented in the S&P 500. However, these firms are among the largest in the country, and they account for about 80% of the market value of traded stocks. Therefore, success for professional investors usually means "beating the S&P."

Some stock market indexes, such as the Wilshire 5000, include an even larger number of stocks, while others focus on special groups of stocks such as the stocks of small companies. There are also stock market indexes for other countries, such as the Nikkei Index for Tokyo and the Financial Times Stock Exchange (FTSE) Index for London. Morgan Stanley Capital International (MSCI) even computes a world stock market index. The Financial Times Company and Standard & Poor's have combined to produce their own world index.

The Historical Record

The historical returns of stock or bond market indexes can give us an idea of the typical performance of different investments. For example, Elroy Dimson, Paul Marsh,

¹ Most stock market indexes record the market value of the portfolio. To calculate the total return on the portfolio, we need to add in any dividends that are paid.

TABLE 11.1 Average rates of return on Treasury bills, government bonds, and common stocks, 1900–2020 (figures in percent per year).

Portfolio	Average Annual Rate of Return, %	Average Premium, % (Extra return versus Treasury bills)
Treasury bills	3.7	
Treasury bonds	5.4	1.7
Common stocks	11.5	7.8

Source: E. Dimson, P. R. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by Triumph's authors.

by themselves. Investors who accepted the risk of common stocks received on average an extra return of 7.8% a year over the return on Treasury bills. This compensation for taking on the risk of common stock ownership is known as the **market risk premium**:

$$\text{Rate of return on common stocks} = \text{interest rate on Treasury bills} + \text{market risk premium}$$

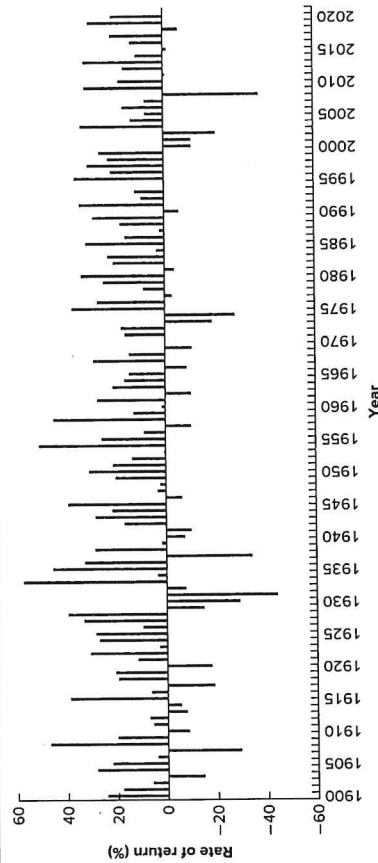
The historical record shows that investors have received a risk premium for holding risky assets. Average returns on high-risk assets have been higher than those on low-risk assets.

You may ask why we look back over such a long period to measure average rates of return. The reason is that annual rates of return for common stocks fluctuate so much that averages taken over short periods are extremely unreliable. In some years, investors in common stocks had a disagreeable shock and received a substantially lower return than they expected. In other years, they had a pleasant surprise and received a higher-than-expected return. By averaging the returns across both the rough years and the smooth, we should get a fair idea of the typical return that investors might justifiably expect.

While common stocks have offered the highest average returns, they have also been riskier investments. Figure 11.2 shows the 121 annual rates of return on common stocks since 1900. The fluctuations in year-to-year returns on common stocks are remarkably wide. There were two years (1933 and 1954) when investors earned a return of more than 50%. However, Figure 11.2 shows that you can also lose money,

risk premium
Expected return in excess of risk-free return as compensation for risk.

FIGURE 11.2 Rates of return on common stocks, 1900–2020.



Source: E. Dimson, P. R. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by Triumph's authors.

and Mike Staunton have compiled measures of the investment performance of three portfolios of securities since 1900:

1. A portfolio of three-month loans issued each week by the U.S. government. These loans are known as *Treasury bills*.
2. A portfolio of long-term *Treasury bonds* issued by the U.S. government and maturing in about 10 years.
3. A diversified portfolio of common stocks.

These portfolios are not equally risky. Treasury bills are about as safe an investment as you can make. Because they are issued by the U.S. government, you can be confident that you will get your money back. Their short-term maturity means that their prices are relatively stable. In fact, investors who wish to lend money for three months can achieve a certain payoff by buying three-month bills. Of course, they can't be sure what that money will buy; there is still some uncertainty about inflation.

Long-term Treasury bonds are also certain to be repaid when they mature, but the prices of these bonds fluctuate more as interest rates vary. When interest rates fall, the values of long-term bonds rise; when rates rise, the values of the bonds fall.

Common stocks are the riskiest of the three groups of securities. When you invest in common stocks, there is no promise that you will get your money back. As a part-owner of the corporation, you receive what is left over after the bonds and any other debts have been repaid.

Figure 11.1 shows the performance of the three groups of securities assuming that all dividends or interest income had been reinvested in the portfolios. You can see that the performance of the portfolios fits our intuitive risk ranking. Common stocks were the riskiest investment, but they also offered the greatest gains. One dollar invested at the start of 1900 in a portfolio of common stocks would have grown to \$69,754 by the start of 2021. At the other end of the spectrum, an investment in Treasury bills would have accumulated to only \$78.

Table 11.1 shows the average of the annual returns from each of these portfolios. These returns are comparable to the return that we calculated for Microsoft. In other words, they include (1) dividends or interest and (2) any capital gains or losses.

The safest investment, Treasury bills, had the lowest rates of return—they averaged 3.7% a year. Long-term government bonds gave slightly higher returns than Treasury bills. This difference is called the **maturity premium**. Common stocks were in a class

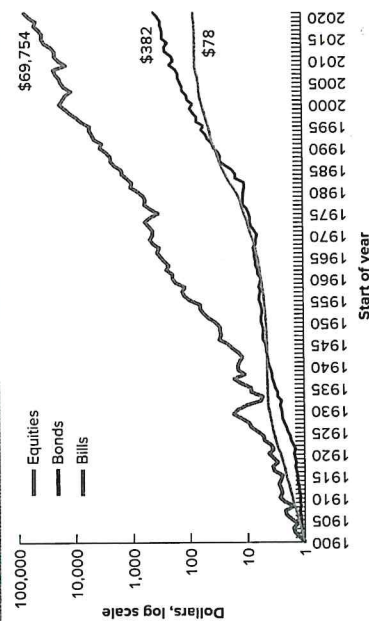


FIGURE 11.1 How an investment of \$1 at the start of 1900 would have grown by the start of 2021 (index values plotted on log scale).

Source: E. Dimson, P. R. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by Triumph's authors.

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Arithmetic averages and compound annual returns



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maturity premium

Extra annual return from investing in long-versus short-term Treasury securities.

and lots of it, by investing in the stock market. The most dramatic case was the stock market crash of 1929–1932. Shortly after President Coolidge joyfully observed that stocks were “cheap at current prices,” stocks rapidly became even cheaper. By July 1932, the Dow Jones Industrial Average had fallen in a series of agonizing slides by 89%.

You don't have to look that far back to see that the stock market is a risky place. Investors who bought at the stock market peak in March 2000 saw little but falling stock prices over the next two and a half years. By October 2002, the S&P 500 had declined by 49%, while the tech-heavy NASDAQ market fell by 78%. But this was not the end of the roller-coaster ride. After recovering sharply, share prices plunged 57% between October 2007 and March 2009 as the financial crisis unfolded.

Bond prices also fluctuate, but far less than stock prices. The worst year for investors in our portfolio of Treasury bonds was 2009; their return that year was –14.9%.

11.2 Self-Test

Here are the average rates of return for common stocks and Treasury bills for four different periods:

	1900–1928	1929–1957	1958–1987	1988–2020
Stocks	12.0%	9.8%	11.8%	12.4%
Treasury bills	4.9	1.0	6.0	3.0

What was the risk premium on stocks for each of these periods?

Using Historical Evidence to Estimate Today's Cost of Capital

Think back now to Chapter 8, where we showed how firms calculate the present value of a new project by discounting the expected cash flows by the opportunity cost of capital. The opportunity cost of capital is the return that the firm's shareholders are giving up by investing in the project rather than in comparable-risk alternatives.

Measuring the cost of capital is easy if the project is a sure thing. Because shareholders can obtain a surefire payoff by investing in U.S. government securities, the firm should invest in a risk-free project only if it can at least match the rate of interest on such an investment. If the project is risky—and most projects are—then the firm needs to at least match the return that shareholders could expect to earn if they invested in securities of similar risk. It is not easy to put a precise figure on this, but our skim through history provides an idea of the average return an investor might expect from an investment in risky common stocks.

Suppose there is an investment project that you *know*—don't ask how—has the same risk as an investment in a diversified portfolio of U.S. common stocks. We will say that it has the same degree of risk as the *market portfolio*.

Instead of investing in the project, your shareholders could invest directly in this market portfolio. Therefore, the opportunity cost of capital for your project is the return that the shareholders could expect to earn on the market portfolio. This is what they are giving up by investing money in your project.

The problem of estimating the project's cost of capital boils down to estimating the currently expected rate of return on the market portfolio. One way to estimate the expected market return is to assume that the future will be like the past and that today's investors expect to receive the average rates of return shown in Table 11.1. In this case, you might judge that the expected market return today is 11.5%, the average of past market returns.

Unfortunately, this is *not* the way to do it. Investors are not likely to demand the same return each year on an investment in common stocks. For example, we know that

the interest rate on safe Treasury bills varies over time. At their peak in 1981, Treasury bills offered a return of 14%, more than 10 percentage points above the 3.7% average return on bills shown in Table 11.1.

What if you were called upon to estimate the expected return on common stocks in 1981? Would you have said 11.5%? That doesn't make sense. Who would invest in the risky stock market for an expected return of 11.5% when you could get a safe 14% from Treasury bills?

A better procedure is to take the *current* interest rate on Treasury bills plus 7.8%, the average risk premium shown in Table 11.1. In 1981, when the rate on Treasury bills was 14%, that would have given

$$\begin{aligned}\text{Expected market} &= \text{interest rate on Treasury bills (1981)} + \text{normal risk premium} \\ \text{return (1981)} &= 14 + 7.8 = 21.8\%\end{aligned}$$

The first term on the right-hand side tells us the time value of money in 1981; the second term measures the compensation for risk. **The expected return on an investment provides compensation to investors both for waiting (the time value of money) and for worrying (the risk of the particular asset).**

What about today? As we write this in early 2021, Treasury bills offer a return of about 0.1%. This suggests that investors in common stocks are looking for a return of 7.9%.²

$$\begin{aligned}\text{Expected market} &= \text{interest rate on Treasury bills (2021)} + \text{normal risk premium} \\ \text{return (2021)} &= 0.1 + 7.8 = 7.9\%\end{aligned}$$

These calculations assume that there is a normal, stable risk premium on the market portfolio, so the expected *future* risk premium can be measured by the average past risk premium. But even with more than 100 years of data, we can't estimate the market risk premium exactly; nor can we be sure that investors today are demanding the same reward for risk that they were in the early 1900s. All this leaves plenty of room for argument about what the risk premium *really* is.

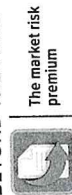
Many financial managers and economists believe that long-run historical returns are the best measure available. Others have a gut instinct that investors don't need such a large risk premium to persuade them to hold common stocks. For example, surveys of financial economists and chief financial officers commonly suggest a risk premium that is 1% to 2% lower than the historical average.³

We may be able to gain some further insights into the question by looking at the experience of other countries. Figure 11.3 shows that the United States is roughly average in terms of the risk premium. Spanish common stocks come bottom of the league; the average risk premium in Spain was only 5.4%. Top of the form is Portugal, with a premium of 10.0%. Some of these variations between countries may reflect differences in risk. But remember how difficult it is to make precise estimates of what investors expected. You probably would not be too far out if you concluded that the *expected* risk premium was the same in each country.

² In practice, things might be a bit more complicated. In 2021, short-term interest rates were relatively low and probably not appropriate for judging the required return on a long-term project. We will return to this problem in the next chapter.

³ For example, a survey of U.S. CFOs in December 2017 produced an average forecast risk premium of 5.7% over the three-month bill rate. A parallel 2020 survey of academics, analysts, and managers likewise found that the average estimate of the required market risk premium for the United States was 5.6%, though this figure seems to represent the premium over a long-term bond rate. See, respectively, Duke/CFO Magazine, “Global Business Outlook Survey,” Fourth Quarter 2020, <http://www.cbssurvey.org/>; and P. Fernandez, E. de Apellaniz, and I. Fernández Achín, “Market Risk Premium and Risk-Free Rate for 81 Countries in 2020: A Survey,” March 25, 2020. Available at SSRN: <https://ssrn.com/abstract=3560809>.

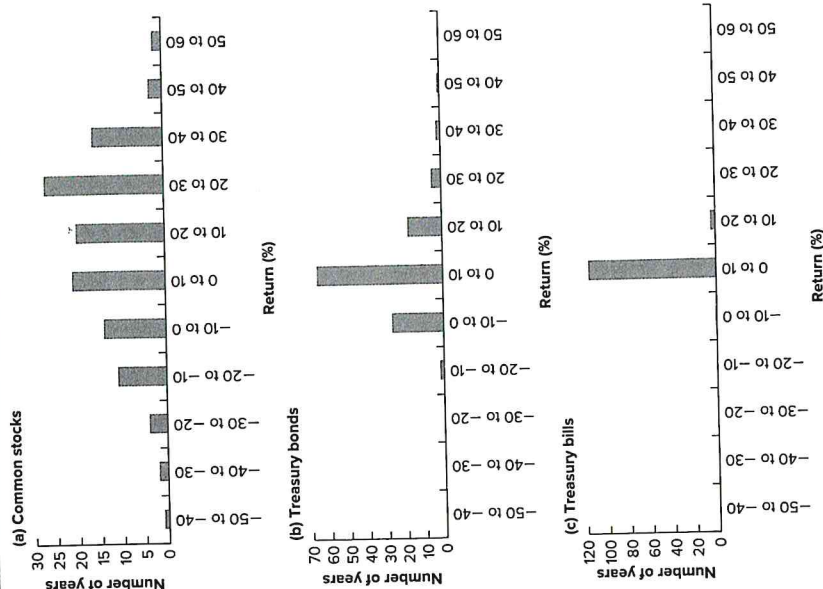
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The market risk premium

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FIGURE 11.4 Historical returns on major asset classes, 1900–2020.



Source: Authors' calculations using data from E. Dimson, P. R. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by Triumph's authors.

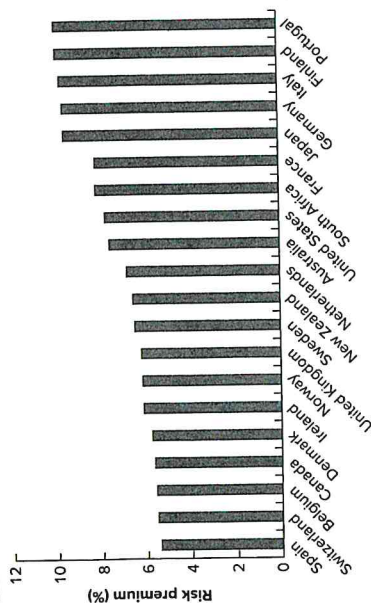
dispersion. But, in general, pictures do not suffice. The financial manager needs a numerical measure of dispersion to quantify risk. The standard measures are **variance** and **standard deviation**.

Here is a very simple example showing how variance and standard deviation are calculated. Suppose that you are offered the opportunity to play the following game: You start by investing \$100. Then two coins are flipped. For each head that comes up, your starting balance will be **increased** by 20%, and for each tail that comes up, your starting balance will be **reduced** by 10%. Clearly, there are four equally likely outcomes:

- Head + Head: You make $20 + 20 = 40\%$
- Head + Tail: You make $20 - 10 = 10\%$
- Tail + Head: You make $-10 + 20 = 10\%$
- Tail + Tail: You make $-10 - 10 = -20\%$

variance
Average value of squared deviations from mean.
A measure of volatility.
standard deviation
Square root of variance.
A measure of volatility.

FIGURE 11.3 The risk premium in 20 countries, 1900–2020. For these 20 countries, the average return on common stocks has been 7.5% more than the interest rate on bills.



Note: The data for Germany omit the hyperinflation years 1922 and 1923.
Source: Authors' calculations using data from E. Dimson, P. R. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by Triumph's authors.

11.3

Measuring Risk

You now have some benchmarks. You know that the opportunity cost of capital for safe projects must be the rate of return offered by safe Treasury bills, and you know that the opportunity cost of capital for "average-risk" projects must be the expected return on the market portfolio. But you *don't* know how to estimate the cost of capital for projects that do not fit these two simple cases. Before you can do this, you need to understand more about investment risk.

The average fuse time for army hand grenades is five seconds, but that average hides a lot of potentially relevant information. If you are in the business of throwing grenades, you need some measure of the variation around the average fuse time.⁴ Similarly, if you are in the business of investing in securities, you need some measure of how far the returns may differ from the average.

One way to present the spread of possible investment returns is by using histograms, such as the ones in Figure 11.4. The bars in each histogram show the number of years between 1900 and 2020 that the investment's return fell within a specific range. Look first at the performance of common stocks. Their risk shows up in the wide spread of outcomes. For example, you can see that in two years, the return was between +50% and +60%, but there was also one year that investors lost between 40% and 50%.

The corresponding histograms for Treasury bonds and bills show that extreme returns, either high or low, are much less common. Investors in these securities could have been much more confident of the outcome than common stockholders.

Variance and Standard Deviation

Investment risk depends on the dispersion or spread of possible outcomes. For example, Figure 11.4 showed that on past evidence, there is greater uncertainty about the possible returns from common stocks than about the returns from bills or bonds. Sometimes, a picture like Figure 11.4 tells you all you need to know about (past)

⁴ We can reassure you: the variation around the standard fuse time is very small.

TABLE 11.2 The coin-toss game; calculating variance and standard deviation.

(1) Rate of Return (%)	(2) Deviation from Expected Return (%)	(3) Squared Deviation
+40	+30	900
+10	0	0
+10	0	0
-20	-30	900

Notes:
1. Variance = average of squared deviations = $1,800/4 = 450$.
2. Standard deviation = square root of variance = $\sqrt{450} = 21.2$, about 21%.

There is a chance of 1 in 4, or .25, that you will make 40%, a chance of 2 in 4, or .5, that you will make 10%; and a chance of 1 in 4, or .25, that you will lose 20%. The game's expected return is, therefore, a weighted average of the possible outcomes:

$$\begin{aligned}\text{Expected return} &= \text{probability-weighted average of possible outcomes} \\ &= (.25 \times 40) + (.5 \times 10) + (.25 \times -20) = +10\%\end{aligned}$$

If you play the game a very large number of times, your average return should be 10%.

Table 11.2 shows how to calculate the variance and standard deviation of the returns on your game. Column 1 shows the four equally likely outcomes. In column 2, we calculate the difference between each possible outcome and the expected outcome. You can see that at best the return could be 30% higher than expected; at worst, it could be 30% lower.

These deviations in column 2 illustrate the spread of possible returns. But if we want a measure of this spread, it is no use just averaging the deviations in column 2—the average is always going to be zero because the positive and negative deviations cancel out. To get around this problem, we *square* the deviations in column 2 before averaging them. These squared deviations are shown in column 3. The variance is the average of these squared deviations and, therefore, is a natural measure of dispersion:

$$\begin{aligned}\text{Variance} &= \text{average of squared deviations around the average} \\ &= \frac{1,800}{4} = 450\end{aligned}\tag{11.2}$$

When we squared the deviations from the expected return, we changed the units of measurement from *percentages* to *percentages squared*. Our last step is to get back to percentages by taking the square root of the variance. This is the standard deviation:

$$\begin{aligned}\text{Standard deviation} &= \text{square root of variance} \\ &= \sqrt{450} = 21\%\end{aligned}\tag{11.3}$$

Because standard deviation is simply the square root of variance, it too is a natural measure of risk. If the outcome of the game had been certain, the standard deviation would have been zero because there would then be no deviations from the expected outcome. The actual standard deviation is positive because we *don't* know what will happen.

Now think of a second game: It is the same as the first except that each head means a 35% gain and each tail means a 25% loss. Again, there are four equally likely outcomes:

- Head + Head: You gain 70%
- Head + Tail: You gain 10%
- Tail + Head: You gain 10%
- Tail + Tail: You lose 50%

The expected return of this game is 10%, the same as that of the first game, but it is riskier. For example, in the first game, the worst possible outcome is a loss of 20%,

which is 30% worse than the expected outcome. In the second game, the downside is a loss of 50%, or 60% below the expected return. This increased spread of outcomes shows up in the standard deviation, which is double that of the first game, 42% versus 21%. By this measure, the second game is twice as risky as the first.

A Note on Calculating Variance

When we calculated variance in Table 11.2, we recorded separately each of the four possible outcomes. An alternative would have been to recognize that in two of the four possible cases the outcomes are the same. In other words, if you were to play the game a large number of times, you would find that on 50% of the occasions the deviation from the expected return is 0%, 25% of the time it is +30%, and the remaining 25% of the time it is -30%. This suggests a simple way to calculate variance. Just weight each squared deviation by its probability:

$$\begin{aligned}\text{Variance} &= \text{sum of squared deviations weighted by probabilities} \\ &= .25 \times 30^2 + .5 \times 0 + .25 \times (-30)^2 = 450\end{aligned}$$

11.3 Self-Test

Calculate the variance and standard deviation of the second (higher-risk) coin-tossing game.

Measuring the Variation in Stock Returns

When estimating the spread of possible outcomes from investing in the stock market, most financial analysts start by assuming that the spread of returns in the past is a reasonable indication of what could happen in the future. Therefore, they calculate the standard deviation of past returns. To illustrate, suppose that you were presented with the data for stock market returns shown in Table 11.3. The average return over the six years from 2015 to 2020 was 12.77%. This is just the sum of the returns over the six years divided by 6 ($76.6/6 = 12.77\%$).

Column 2 in Table 11.3 shows the difference between each year's return and the average return. For example, in 2017 the return of 21.0% on common stocks was above

TABLE 11.3 The average return and standard deviation of stock market returns, 2015–2020.

Year	Rate of Return (%)	Deviation from Average Return (%)	Squared Deviation
2015	-1.6	-14.3	205.75
2016	13.0	0.2	0.04
2017	21.0	8.2	66.96
2018	-6.1	-18.9	357.88
2019	29.8	17.0	289.94
2020	20.6	7.8	60.76
Total	76.6		981.32
Average return = $76.6/6 = 12.77\%$			
Variance = average of squared deviations = $981.32/6 = 163.55$			
Standard deviation = square root of variance = 12.79%			

Source: Authors' calculations using data from E. Dimson, P. R. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by Triumph's authors.

TABLE 11.4 Standard deviation of returns, 1900–2020.

Portfolio	Standard Deviation (%)
Treasury bills	2.8
Long-term government bonds	8.9
Common stocks	19.5

Source: Authors' calculations using data from E. Dimson, P. R. Marsh, and M. Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns* (Princeton, NJ: Princeton University Press, 2002), with updates kindly provided by Triumph's authors.

the six-year average by 8.2%. In column 3, we square these deviations from the average. The variance is then the average of these squared deviations.⁵

Variance = average of squared deviations

$$= \frac{981.32}{6} = 163.55$$

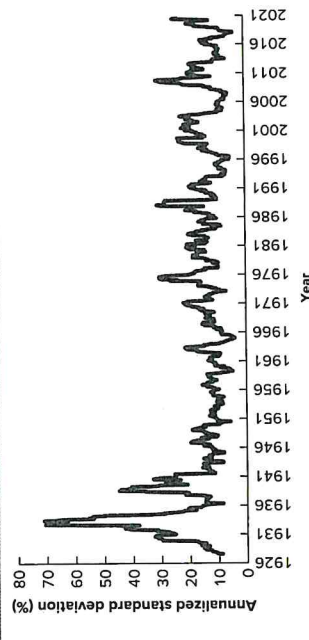
Because standard deviation is the square root of the variance,

$$\begin{aligned}\text{Standard deviation} &= \text{square root of variance} \\ &= \sqrt{163.55} = 12.79\%\end{aligned}$$

We can't reliably estimate risk on the basis of just six past outcomes. Therefore, Table 11.4 lists the annual standard deviations for our three portfolios of securities over the period 1900–2020. As expected, Treasury bills were the least variable security, and common stocks were the most variable. Treasury bonds hold the middle ground.

Of course, there is no reason to believe that the market's variability should stay the same over many years. Indeed many people believe that, in recent years, the stock market has become more volatile due to irresponsible speculation by ... (fill in here the name of your preferred guilty party). Figure 11.5 provides a chart of the volatility of the U.S. stock market for each year from 1926 to 2020.⁶ Notice how volatility spiked upward

FIGURE 11.5 Annualized standard deviation of monthly excess returns on U.S. equities.



Source: Authors' calculations, using data from Prof. Kenneth French's web site, https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

⁵ Technical note: When variance is estimated from a sample of observed returns, it is common to add the squared deviations and divide by $N - 1$, rather than N , where N is the number of observations. This procedure adjusts the estimate for what is called *the loss of a degree of freedom*. We will ignore this fine point, emphasizing the interpretation of variance as an average squared deviation. In any event, the correction for the loss of freedom is negligible when there are plentiful observations. For example, with 100 years of data, the difference between dividing by 99 or 100 will affect the estimated variance by only 1% (i.e., a multiple of 1.01).

⁶ We converted the weekly variance to an annual variance by multiplying by 52. In other words, the variance of annual returns is 52 times that of weekly returns. The longer you hold a security, the more risk you incur.

11.4

Risk and Diversification

Diversification

We can calculate our measures of variability equally well for individual securities and portfolios of securities. Of course, the level of variability over 100 years is less interesting for specific companies than for the market portfolio because it is a rare company that faces the same business risks today as it did a century ago.

Table 11.5 presents estimated standard deviations for some well-known common stocks for a recent five-year period.⁷ The standard deviation of the market index (the S&P 500) in these years was 12.0%, well below the long-term average. However, the standard deviation of the returns on the individual stocks was much higher than 12.0%.

This raises an important question: The market portfolio is made up of individual stocks, so why isn't its variability equal to the average variability of its components? The answer is that **diversification reduces variability**.

Selling umbrellas is a risky business; you may make a killing when it rains, but you are likely to lose your shirt in a heat wave. Selling ice cream is no safer: you do well in the heat wave, but business is poor in the rain. Suppose, however, that you invest in both an umbrella shop and an ice cream shop. By diversifying your investment across the two businesses, you make an average level of profit come rain or shine.

diversification
Strategy designed to
reduce risk by spreading
the portfolio across many
investments.

TABLE 11.5 Standard deviations for selected common stocks, January 2015–December 2019.

Ticker	Company	Standard deviation (%)
X	U.S. Steel	76.4
PCG	Pacific Gas & Electric	55.4
MRO	Marathon Oil	48.9
NEM	Newmont Mining	36.7
AMZN	Amazon	28.3
BA	Boeing	27.4
CPB	Campbell Soup	22.6
IBM	IBM	22.5
F	Ford	22.1
INTC	Intel	22.1
UNP	Union Pacific	21.4
DIS	Disney	21.2
GOOG	Google/Alphabet	20.1
SBUX	Starbucks	18.7
WMT	Walmart	17.7
XOM	ExxonMobil	19.4
PFE	Pfizer	16.6
MCD	McDonald's	14.4
KO	Coca-Cola	12.6
S&P 500		12.0

⁷ We pointed out earlier that five annual observations are insufficient to give a reliable estimate of variability. Therefore, these estimates are derived from 60 monthly rates of return, and then the monthly variance is multiplied by 12.

Portfolio diversification works because prices of different stocks do not move exactly together. Statisticians make the same point when they say that stock price changes are less than perfectly correlated. Diversification works best when the returns are negatively correlated, as is the case of our umbrella and ice cream businesses. When one business does well, the other does badly. Unfortunately, in practice, stocks that are negatively correlated are as rare as a summer snowstorm.

Asset versus Portfolio Risk

The history of returns on different asset classes provides compelling evidence of a risk-return trade-off and suggests that the variability of the rate of return on a broad asset class is a useful measure of risk. However, volatility of returns can be a misleading measure of risk for an individual asset held as part of a portfolio. To see why, consider the following example.

Suppose there are three equally likely outcomes, or *scenarios*, for the economy: a recession, normal growth, and a boom. An investment in an auto stock will have a rate of return of -8% in a recession, 5% in a normal period, and 18% in a boom. Auto firms are *cyclical*. They do well when the economy does well. In contrast, gold firms are often said to be *countercyclical*, meaning that they do well when other firms do poorly. Suppose that stock in a gold mining firm will provide a rate of return of 20% in a recession, 3% in a normal period, and -20% in a boom. These assumptions are summarized in Table 11.6.

It appears that gold is the more volatile investment. The difference in return across the boom and bust scenarios is 40% (-20% in a boom versus +20% in a recession), compared with a spread of only 26% for the auto stock. In fact, we can confirm the higher volatility by measuring the variance or standard deviation of returns of the two assets. The calculations are set out in Table 11.7.

TABLE 11.6 Rate of return assumptions for two stocks.

Scenario	Probability	Rate of Return (%)	
		Auto Stock	Gold Stock
Recession	1/3	-8	+20
Normal	1/3	+5	+3
Boom	1/3	+18	-20

TABLE 11.7 Expected return and volatility for two stocks.

Scenario	Auto Stock		Gold Stock	
	Rate of Return (%)	Deviation from Expected Return (%)	Rate of Return (%)	Deviation from Expected Return (%)
Recession	-8	-13	+20	+19
Normal	+5	0	+3	+2
Boom	+18	+13	-20	-21
Expected return	$\frac{1}{3}(-8 + 5 + 18) = 5\%$		$\frac{1}{3}(+20 + 3 - 20) = 1\%$	
Variance*	$\frac{1}{3}(169 + 0 + 169) = 112.7$		$\frac{1}{3}(361 + 4 + 441) = 268.7$	
Standard deviation (= $\sqrt{\text{variance}}$)	$\sqrt{112.7} = 10.6\%$		$\sqrt{268.7} = 16.4\%$	

*Variance = average of squared deviations from the expected value.

11.4 Self-Test

Suppose the probabilities of the recession or boom are each .30, while the probability of a normal period is .40. Would you expect the variance of returns on these two investments to be higher or lower? Why? Confirm by calculating the standard deviation of the auto stock. (Refer to Section 11.3 if you are unsure of how to do this.)

Because all three scenarios are equally likely, the expected return on each stock is simply the average of the three possible outcomes.⁸ For the auto stock, the expected return is 5% for the gold stock, it is 1%. The variance is the average of the squared deviations from the expected return, and the standard deviation is the square root of the variance.

The gold mining stock offers a lower expected rate of return than the auto stock and *more* volatility—a loser on both counts, right? Would anyone be willing to hold gold mining stocks in an investment portfolio? The answer is a resounding yes.

To see why, suppose you do believe that gold is a lousy asset, and therefore, you hold your entire portfolio in the auto stock. Your expected return is 5% and your standard deviation is 10.6%. We'll compare that portfolio to a partially diversified one, invested 75% in autos and 25% in gold. For example, if you have a \$10,000 portfolio, you could put \$7,500 in autos and \$2,500 in gold.

First, we need to calculate the return on this portfolio in each scenario. The portfolio return is the weighted average of returns on the individual assets with weights equal to the proportion of the portfolio invested in each asset. For a portfolio formed from only two assets,

$$\text{Portfolio rate of return} = \left(\frac{\text{fraction of portfolio on first asset}}{\text{rate of return}} \right) + \left(\frac{\text{fraction of portfolio on second asset}}{\text{rate of return}} \right) \quad (11.4)$$

For example, autos have a weight of .75 and a rate of return of -8% in the recession, and gold has a weight of .25 and a return of 20% in a recession. Therefore, the portfolio return in the recession is the following weighted average:⁹

$$\text{Portfolio return in recession} = [.75 \times (-8\%)] + (.25 \times 20\%) = -1\%$$

Table 11.8 expands Table 11.6 to include the portfolio of the auto stock and the gold mining stock. The expected returns and volatility measures are summarized at the bottom of the table. The surprising finding is this: When you shift part of your funds from the auto stock to the more volatile gold mining stock, your portfolio variability actually *decreases*. In fact, the volatility of the auto-plus-gold stock portfolio is considerably less than the volatility of either stock separately. This is the payoff to diversification.

We can understand this more clearly by focusing on asset returns in the two extreme scenarios, boom and recession. In the boom, when auto stocks do best, the poor return on gold reduces the performance of the overall portfolio. However, when auto stocks are stalling in a recession, gold shines, providing a substantial positive return that boosts portfolio performance. The gold stock offsets the swings in the performance of

⁸ If the probabilities were not equal, we would need to weight each outcome by its probability in calculating the expected outcome and the variance.

⁹ Let's confirm this. Suppose you invest \$7,500 in autos and \$2,500 in gold. If the recession hits, the rate of return on autos will be -8%, and the value of the auto investment will fall by 8% to \$6,900. The rate of return on gold will be 20%, and the value of the gold investment will rise 20% to \$3,000. The value of the total portfolio falls from its original value of \$10,000 to \$6,900 + \$3,000 = \$9,900, which is a rate of return of -1%. This matches the rate of return given by the formula for the weighted average.

TABLE 11.8 Rates of return for two stocks and a portfolio with 75% invested in the auto stock and 25% in the gold stock.

Scenario	Probability	Auto Stock	Gold Stock	Portfolio Return (%)
Recession	1/3	-8	+20	-1.0
Normal	1/3	+5	+3	+4.5
Boom	1/3	+18	-20	+8.5
Expected return		5	1	4
Variance		112.7	268.7	15.2
Standard deviation		10.6	16.4	3.9

* Portfolio return = $(75 \times \text{auto stock return}) + (25 \times \text{gold stock return})$.

the auto stock, reducing the best-case return but improving the worst-case return. The inverse relationship between the returns on the two stocks means that returns are more stable when the gold mining stock is added to an all-auto portfolio.

A gold stock is actually a *negative-risk* asset to an investor starting with an all-auto portfolio. Adding it to the portfolio reduces the volatility of returns. The *incremental* risk of the gold stock (i.e., the *change* in overall risk when gold is added to the portfolio) is *negative* despite the fact that gold returns are highly volatile.

In Table 11.9, we consider several other potential portfolios, all formed by mixing our gold and auto stocks in varying proportions. Portfolio A is invested fully in the auto stock, portfolio B shifts 20% of the portfolio from the auto stock to gold, and so on, until we reach portfolio F, which is fully invested in gold. The table shows the rate of return of each portfolio in each scenario; the expected value and standard deviation of returns across the three scenarios appear in the last two columns. Notice that the standard deviation of portfolio A is 10.6%, which of course is just the standard deviation of the auto stock. When we shift 20% of the portfolio to more volatile gold, as in portfolio B, standard deviation actually falls. As we've seen, this is the benefit of diversification.

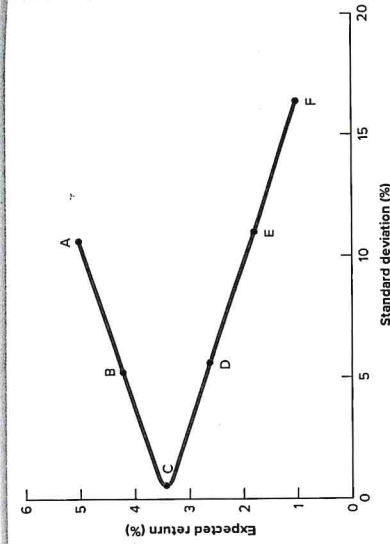
How much more can we reduce risk? The standard deviation of portfolio C, which has a 40% weight in gold, is even lower. But this is about the best we can do. Beyond this point, adding more gold *increases* standard deviation, to 5.6% for portfolio D and 11% for portfolio E. These portfolios are already heavily invested in gold, so adding more of it increases risk. Thus, the incremental risk of gold depends on where you are starting from. Portfolios A and B are dominated by the auto stock, so adding gold *reduces* volatility. But portfolios D and E are already dominated by gold, so adding more now *increases* volatility.

Figure 11.6 plots the expected return–standard deviation pairs of our six portfolios. The “extreme” portfolios, A and F, which are fully invested in either autos or gold, are at the two ends of the plot. When we “connect the dots” corresponding to each portfolio, we trace out the possible combinations of expected return and portfolio risk. This plot is called the **investment opportunity frontier**. The frontier dramatically illustrates the benefit of diversification. In our example, as portfolio C shows, risk can be driven almost to zero. The great power of diversification in this case derives from the strong inverse relation between the gold and the auto stocks. If the relationship were less strong, the investment opportunity frontier would have the same general shape, but it would not come so close to the vertical axis.

	Portfolio Weights				Portfolio Rate of Return (%)			Expected Return	Standard Deviation
	Gold	Autos	Recession	Normal	Boom				
A	0.0	1.0	-8.0	5.0	18.0	5.0	10.6		
B	0.2	0.8	-2.4	4.6	10.4	4.2	5.2		
C	0.4	0.6	3.2	4.2	2.8	3.4	0.6		
D	0.6	0.4	8.8	3.8	-4.8	2.6	5.6		
E	0.8	0.2	14.4	3.4	-12.4	1.8	11.0		
F	1.0	0.0	20.0	3.0	-20.0	1.0	16.4		

TABLE 11.9 Risk and return on portfolios formed by mixing the auto and the gold stocks in varying proportions.

FIGURE 11.6 The investment opportunity frontier for the gold and auto stocks. Each point on the curve represents a feasible combination of expected return and volatility. The six labeled points correspond to the portfolios in Table 11.9.



In general, the incremental risk of a stock depends on whether its returns tend to vary with or against the returns of the other assets in the portfolio. Incremental risk does not just depend on a stock's volatility. If returns do not move closely with those of the rest of the portfolio, the stock generally will reduce the volatility of portfolio returns. The degree to which two stocks move together can be measured by the *correlation* between their returns. If the gold and auto stocks in Table 11.7 moved in perfect lockstep, the correlation would be 1.0. If their returns were completely unrelated, the correlation would be zero. Because their returns actually move inversely, that is, one stock goes up when the other goes down, the correlation is negative. The lowest possible correlation is -1.0 , which indicates that returns move in perfect lockstep but in opposite directions. In our example, the correlation between the gold and auto stocks was nearly this extreme at -996 . Unfortunately, in practice, negative correlations are rare because most stocks have a common dependence on the overall economy.

Table 11.10 shows correlations across a few major industries calculated from five years of monthly stock returns ending in 2020. You can find the correlation between any industry pair by picking off the number in the relevant row and column. Of course, each industry is perfectly correlated with itself, so every entry on the diagonal is exactly 1.

As we would expect, most correlations in the table are positive, but the correlations are highest between pairs of industries that are very sensitive to the business cycle. The maximum correlation in the table, .78, is between the construction and auto industries. Correlations between these industries and less “cyclical” industries such as gold

TABLE 11.10 Correlations across some major industries.

	Food	Drugs	Construct	Steel	Autos	Gold	Oil	Utilities	Telecom	Retail	Banks
Food	1.00										
Drugs	0.46	1.00									
Construction	0.50	0.52	1.00								
Steel	0.37	0.59	0.62	1.00							
Autos	0.46	0.58	0.78	0.68	1.00						
Gold	0.26	0.22	0.10	0.30	0.28	1.00					
Oil	0.51	0.54	0.75	0.73	0.71	0.34	1.00				
Utilities	0.61	0.30	0.58	0.30	0.38	0.23	0.50	1.00			
Telecom	0.69	0.62	0.66	0.61	0.61	0.26	0.69	0.60	1.00		
Retail	0.65	0.71	0.66	0.57	0.66	0.13	0.39	0.66	1.00		
Banks	0.46	0.63	0.78	0.73	0.76	-0.04	0.76	0.68	0.65	1.00	

Source: Authors' calculations using monthly value-weighted industry index returns for the five-year period ending June 2020, downloaded from the Fama-French data library: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_10_ind_port.html.

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Expected return and risk of a 2-stock portfolio



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Investment opportunity frontier

Plot of the combinations of expected return versus standard deviation for various portfolio weights.

Spreadsheet Solutions Calculating Volatility

Excel and most other spreadsheet programs provide built-in functions for calculating standard deviation and correlation. In columns B and C of the following spreadsheet, we have entered returns for the S&P 500 and Alphabet (the parent company of Google) for six months in 2020. In practice, estimates based on only six months of data would be very unreliable, but our goal here is simply to illustrate the technique. Real-world estimates would be more likely to use 60 monthly returns or perhaps 52 weekly returns.

Here are some points to notice about the spreadsheet:

- 1. Columns B and C. These columns show monthly returns for the S&P 500 and Alphabet (GOOG). Sometimes people mistakenly enter prices instead of returns and get nonsensical results.
- 2. Row 11. Footnote 5 pointed out that in estimating standard deviation from a sample of observations, it is common to make an adjustment for the loss of a "degree of freedom." To do this, we would use the Excel formula STDEV rather than STDEV.P. In some versions of Excel, the formulas are STDEV.S (which corrects for degrees of freedom) and STDEV.P (which does not).

	A	B	C	D
1				
2		Returns (%)	Formula Used in	
3	Month	S&P 500	GOOG	Column C
4	Jul-20	7.01%	10.20%	
5	Aug-20	-3.92%	-10.07%	
6	Sep-20	-2.77%	10.30%	
7	Oct-20	10.75%	8.62%	
8	Nov-20	3.71%	-0.50%	
9	Dec-20	-0.16%	6.97%	
10	Mean return	2.44%	4.25%	=AVERAGE(C4:C9)
11	Standard deviation (monthly)	5.26%	7.37%	=STDEV.P(C4:C9)
12	Standard deviation (annualized)	18.22%	25.54%	=C12*SQRT(12)
13	Correlation		0.47	=CORREL(C4:C9,B4:B9)

mining are considerably lower. The real-life correlation between gold and autos is .28, quite low, but still not negative as the value used in our example.

Excel provides built-in formulas that make calculating correlations as well as standard deviations pretty easy. We show you how in the nearby box.

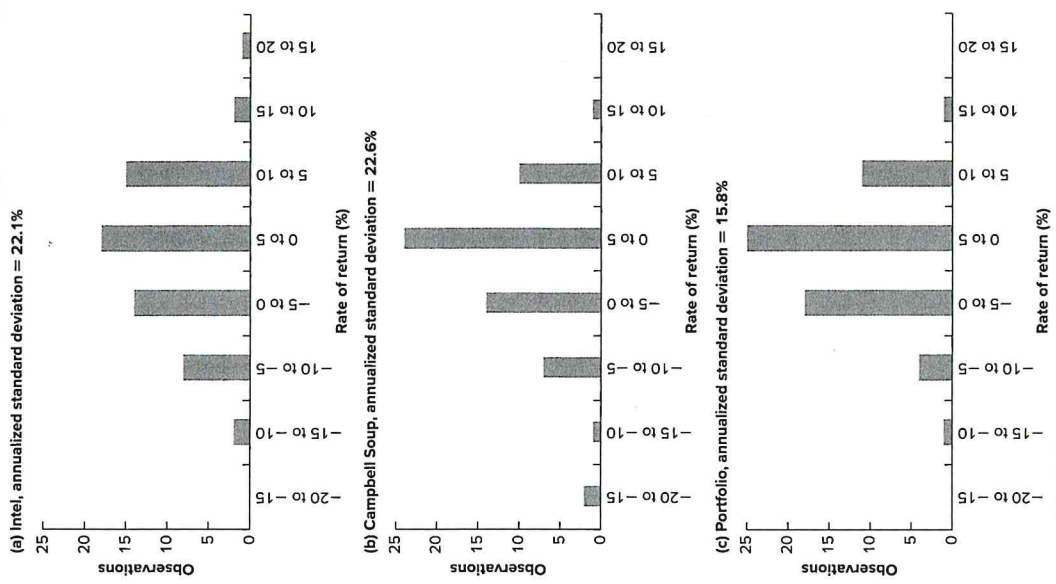
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How to calculate correlation coefficients

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FIGURE 11.7 The rate of return of a portfolio evenly divided between Intel and Campbell Soup was less volatile than either stock on its own.



from diversification depends on the correlations between stocks in the portfolio. Portfolios of stocks all taken from one industry, for example, would not benefit much from diversification because the returns would be highly correlated. A portfolio diversified across different industries would benefit more because correlations would be lower.

Example 11.1 ► Diversification

Our example of the auto and gold mining stocks was entirely fanciful, but we can make the same point by looking at two real firms, Intel and Campbell Soup. Panels a and b of Figure 11.7 show the histogram of monthly returns on the two stocks for the five-year period ending December 2019. Although both stocks had a wide spread of monthly returns, they did not move in exact lockstep. Often, a decline in the value of one stock was offset by a rise in the price of the other. Panel c shows that if you had split your portfolio equally between the two stocks, you would have reduced the variation in the returns on your investment. The standard deviation of the combined portfolio would have been about a third less than the average risk of the two individual stocks. ■

11.5 Self-Test

An investor is currently fully invested in gold mining stocks. Which action would do more to reduce portfolio risk: diversification into silver mining stocks or into automotive stocks? Why?

Market Risk versus Specific Risk

Our examples illustrate that even a little diversification can provide a substantial reduction in variability. Suppose you calculate and compare the standard deviations of randomly chosen one-stock portfolios, two-stock portfolios, five-stock portfolios, and so on. You can see from Figure 11.8 that diversification can cut the variability of returns by about half. But you can get most of this benefit with relatively few stocks: The improvement is slight when the number of stocks is increased beyond, say, 20 or 30.

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How diversification reduces risk

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FIGURE 11.8 The risk (standard deviation) of portfolios containing different numbers of New York Stock Exchange stocks. The stocks were selected randomly from New York Stock Exchange stocks from 2010 through 2019. Notice that diversification reduces risk rapidly at first and then more slowly.

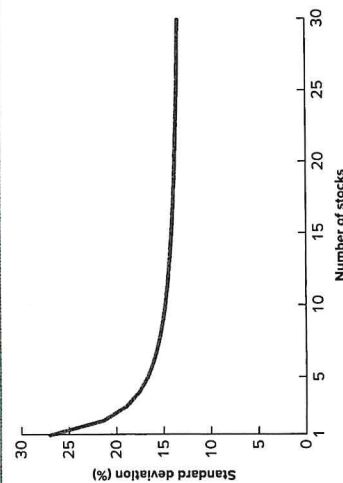
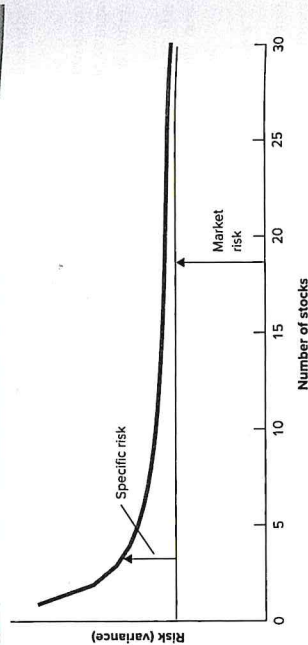


FIGURE 11.9 Diversification eliminates specific risk. But there is some risk that diversification cannot eliminate. This is called market risk.



Technical note: Risk here is measured by the variance. The total variance of a portfolio is the sum of the variance due to the market and the specific variance.

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International diversification

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specific risk
Risk factors affecting only that firm. Also called *diversifiable risk*.

market risk
Economywide (macroeconomic) sources of risk that affect the overall stock market. Also called *systematic risk*.

Figure 11.8 also illustrates that no matter how many securities you hold, you cannot eliminate all risk. If the economy as a whole tanks, then most stocks, and even diversified portfolios, are likely to plummet.

The risk that can be eliminated by diversification is called **specific risk**. The risk that you can't avoid regardless of how much you diversify is generally known as **market risk** or **systematic risk**. *Specific risk* arises because many of the perils that surround an individual company are peculiar to that company and perhaps its direct competitors. *Market risk* stems from economywide perils that threaten all businesses. Market risk explains why stocks have a tendency to move together, so even well-diversified portfolios are exposed to market movements.

Figure 11.9 divides risk into its two parts—specific risk and market risk. If you have only a single stock, specific risk is very important; but once you have a portfolio of 30 or more stocks, diversification has done most of what it can to eliminate risk. For a reasonably well-diversified portfolio, only market risk matters.

11.5 Thinking about Risk

How can you tell which risks are specific and diversifiable? Where do market risks come from? Here are three messages to help you think clearly about risk.

Message 1: Some Risks Look Big and Dangerous but Really Are Diversifiable

Managers confront risks “up close and personal.” They must make decisions about particular investments. The failure of such an investment could cost a promotion, bonus, or otherwise steady job. Yet that same investment may not seem risky to an investor who can stand back and combine it in a diversified portfolio with many other assets or securities.

Example 11.2 ► Wildcat Oil Wells

You have just been promoted to director of exploration, Western Hemisphere, of MPS Oil. The manager of your exploration team in far-off Costaguana has appealed for \$20 million extra to drill in an even steamier part of the Costaguanan jungle. The manager thinks there

may be an "elephant" field worth \$500 million or more hidden there. But the chance of finding it is at best 1 in 10, and yesterday MPS's CEO sourly commented on the \$100 million already "wasted" on Costaguanan exploration.

Is this a risky investment? For you it probably is; you may be a hero if oil is found and a goat otherwise. But MPS drills hundreds of wells worldwide; for the company as a whole, it's the average success rate that matters. Geologic risks (is there oil or not?) should average out. The risk of a worldwide drilling program is much less than the apparent risk of any single wildcat well.

Back up one step, and think of the investors who buy MPS stock. The investors may hold other oil companies, too, as well as companies producing steel, computers, clothing, cement, and breakfast cereal. They naturally—and realistically—assume that your successes and failures in drilling oil wells will average out with the thousands of other bets made by the companies in their portfolio.

Therefore, the risks you face in Costaguana do not affect the rate of return investors demand for holding the stock of MPS Oil. Diversified investors in MPS stock will be happy if you find that elephant field, but they probably will not notice if you fail and lose your job. In any case, they will not demand a higher average rate of return for worrying about geologic risks in Costaguana. ■

Example 11.3 ▶

Fire Insurance

Would you be willing to write a \$100,000 fire insurance policy on your neighbor's house? The neighbor is willing to pay you \$100 for a year's protection, and experience shows that the chance of fire damage in a given year is substantially less than 1 in 1,000. But if your neighbor's house is damaged by fire, you would have to pay up.

Few of us have deep enough pockets to insure our neighbors, even if the odds of fire damage are very low. Insurance seems a risky business if you think policy by policy. But a large insurance company, which may issue a million policies, is concerned only with average losses, which can be predicted with excellent accuracy. ■

11.6 Self-Test

Imagine a laboratory at Intel, late at night. One scientist speaks to another.

"You're right, Watson, I admit this experiment will consume all the rest of this year's budget. I don't know what we'll do if it fails. But if this yttrium-magnesium alloy superconducts, the patents will be worth millions."

Would this be a good or bad investment for Intel? Can't say. But from the ultimate investors' viewpoint this is *not* a risky investment. Explain why.

Message 2: Market Risks Are Macro Risks

We have seen that diversified portfolios are not exposed to the specific risks of individual stocks but are exposed to the uncertain events that affect the entire securities market and the entire economy. These are macroeconomic, or "macro," factors such as changes in interest rates, industrial production, inflation, foreign exchange rates, and energy costs. These factors affect most firms' earnings and stock prices. When the relevant macro risks turn generally favorable, stock prices rise and investors do well; when the same variables go the other way, investors suffer.

You can often assess relative market risks just by thinking through exposures to the business cycle and other macro variables. The following businesses have substantial macro and market risks:

- *Airlines.* Because business travel falls during a recession, and individuals postpone vacations and other discretionary travel, the airline industry is subject to the swings of the business cycle. On the positive side, airline profits really take off when business is booming and personal incomes are rising.
- *Machine tool manufacturers.* These businesses are especially exposed to the business cycle. Manufacturing companies that have excess capacity rarely buy new machine tools to expand. During recessions, excess capacity can be quite high.

Here, on the other hand, are two industries with less-than-average macro exposures:

- *Food companies.* Companies selling staples, such as breakfast cereal, flour, and dog food, find that demand for their products is relatively stable in good times and bad.
- *Electric utilities.* Business demand for electric power varies somewhat across the business cycle, but by much less than demand for air travel or machine tools. Also, many electric utilities' profits are regulated. Regulation cuts off upside profit potential but also gives the utilities the opportunity to increase prices when demand is slack.

Remember, investors holding diversified portfolios are mostly concerned with macroeconomic risks. They do not worry about microeconomic risks peculiar to a particular company or investment project. Micro risks wash out in diversified portfolios. Company managers may worry about both macro and micro risks, but only the former affect diversified investors and the cost of capital.

11.7 Self-Test

Which company of each of the following pairs would you expect to be more exposed to macro risks?

- A luxury Manhattan restaurant or an established Burger Queen franchise?
- A paint company that sells through small paint and hardware stores to do-it-yourselfers or a paint company that sells in large volumes to Ford, GM, and Chrysler?

Message 3: Risk Can Be Measured

Delta Airlines clearly has more exposure to macro risks than food companies such as Kellogg or General Mills. These are easy cases. But is Intel stock a riskier investment than ExxonMobil? That's not an easy question to reason through. We can, however, *measure* the risk of Intel and ExxonMobil by looking at how their stock prices fluctuate.

We've already hinted at how to do this. Remember that diversified investors are concerned with market risks. The movements of the stock market sum up the net effects of all relevant macroeconomic uncertainties. If the market portfolio of all traded stocks is up in a particular month, we conclude that the net effect of macroeconomic news is positive. Remember, the performance of the market is barely affected by a firm-specific event. These cancel out across thousands of stocks in the market.

How do we measure the risk of a single stock like Intel or ExxonMobil? We do not look at the stocks in isolation because the risks that loom when you're up close to a single company are often diversifiable. Instead, we measure the individual stock's sensitivity to the fluctuations of the overall stock market. We will show you how this works in the next chapter.

SUMMARY

How can one estimate the opportunity cost of capital for an "average-risk" project? (LO11-1)

- Over the past century, the average return on the Standard & Poor's Composite Index of common stocks has been 7.8% a year higher than the return on safe Treasury bills. This is the **risk premium** that investors have received for taking on the risk of investing in stocks.
- If the historical risk premium is a guide to the future, we can estimate the expected return on the market today by adding that 7.8% expected risk premium to today's interest rate on Treasury bills.
- The expected return on the market would be the opportunity cost of capital for an average-risk project—that is, one with the same risk as the broad market index.

How is the standard deviation of returns for individual common stocks or for a stock portfolio calculated? (LO11-2)

- The spread of outcomes on different investments is commonly measured by the **variance** or **standard deviation** of the possible outcomes. The variance is the average of the squared deviations around the average outcome, and the standard deviation is the square root of the variance.
- The standard deviation of the returns on a market portfolio of common stocks has averaged around 20% a year over the last century.

Why does diversification reduce risk? (LO11-3)

- Because individual stocks do not move in exact lockstep, much of their risk can be diversified away, with good or bad firm-specific outcomes canceling out across the many stocks in a diversified portfolio.
- By spreading your portfolio across many investments, you smooth out the risk of your overall position. The risk that can be eliminated through diversification is known as **specific risk**.
- Because of diversification, the standard deviation of the broad market is generally lower than the standard deviation of individual stock returns.

What is the difference between specific risk, which can be diversified away, and market risk, which cannot? (LO11-4)

- Risks that are specific to the firm will be largely diversified away in large portfolios.
- But even if you hold a highly diversified portfolio, you will not eliminate all risk because there are some risks that affect pretty much all stocks in tandem. These tend to be risks that affect the entire macroeconomy. Because they affect most stocks simultaneously, these macro risks result in **market risk**—that is, the risk that the market as a whole will slump.
- A "high-risk stock" is not necessarily one with high total volatility because much of that risk may be diversifiable. Rather, a high-risk stock is one that makes an above-average contribution to the risk of a diversified portfolio. This contribution depends on the stock's sensitivity to macroeconomic conditions or, equivalently, on its market risk.
- Investors shouldn't worry much about the risk that they can diversify away; they *do* need to worry about the risk that cannot be diversified.

LISTING OF EQUATIONS

- 11.1 $\text{Percentage return} = \frac{\text{capital gain} + \text{dividend}}{\text{initial share price}}$
- 11.2 $\text{Variance} = \text{average of squared deviations from the average}$
- 11.3 $\text{Standard deviation} = \text{square root of variance}$
- 11.4 $\text{Portfolio rate of return} = \left(\frac{\text{fraction of portfolio in first asset}}{\text{in first asset}} \times \text{rate of return on first asset} \right) + \left(\frac{\text{fraction of portfolio in second asset}}{\text{in second asset}} \times \text{rate of return on second asset} \right)$

QUESTIONS AND PROBLEMS



- Stock Market History.** Use the data in Tables 11.1 and 11.4 to answer these questions: (LO11-1)
 - What was the average rate of return on large U.S. common stocks from 1900 to 2020?
 - What was the average risk premium on large stocks?
 - What was the standard deviation of returns on the market portfolio?
- Maturity Premiums.** Investments in long-term government bonds produced a negative average return during the period 1977–1981. How should we interpret this? Did bond investors in 1977 expect to earn a negative maturity premium? What do these five years of bond returns tell us about the normal future maturity premium? (LO11-1)
- Risk Premiums.** What will happen to the opportunity cost of capital if investors suddenly become especially conservative and less willing to bear investment risk? (LO11-1)
- Risk Premium.** If the stock market return next year turns out to be –20%, will our estimate of the "normal" risk premium increase or decrease? Does this make sense? (LO11-1)
- Risk Premiums and Discount Rates.** Top hedge fund manager Sally Buffit believes that a stock with the same market risk as the S&P 500 will sell at year-end at a price of \$50. The stock will pay a dividend at year-end of \$2. What price should she be willing to pay for the stock today? Assume that risk-free Treasury securities currently offer an interest rate of 2%. Use Table 11.1 to find a reasonable discount rate. (LO11-1)
- Risk Premiums.** Here are rates of return on a broad stock market index and on Treasury bills between 2016 and 2020: (LO11-1)

Year	Stock Market Return (%)	T-Bill Return (%)
2016	13.0	0.2
2017	21.0	0.8
2018	–6.2	1.8
2019	29.8	2.1
2020	20.6	0.4

- What was the risk premium on common stock in each year?
 - What was the average risk premium?
 - What was the standard deviation of the risk premium?
- Rate of Return.** A stock is selling today for \$40 per share. At the end of the year, it pays a dividend of \$2 per share and sells for \$44. (LO11-2)
 - What is the total rate of return on the stock?
 - What are the dividend yield and percentage capital gain?
 - Now suppose the year-end stock price after the dividend is paid is \$36. What are the dividend yield and percentage capital gain in this case?
 - Real versus Nominal Returns.** You purchase 100 shares of stock for \$40 a share. The stock pays a \$2 per share dividend at year-end. (LO11-2)
 - What is the rate of return on your investment if the end-of-year stock price is (i) \$38; (ii) \$40; (iii) \$42?
 - What is your real (inflation-adjusted) rate of return if the inflation rate is 4%?
 - Real versus Nominal Returns.** The Costaguanan stock market provided a rate of return of 95%. The inflation rate in Costaguanan during the year was 80%. In Ruritania, in contrast, the stock market return was only 12%, but the inflation rate was only 2%. Which country's stock market provided the higher real rate of return? (LO11-2)
 - Real versus Nominal Returns.** The inflation rate in the United States has averaged about 3% a year since 1900. What was the average real rate of return on Treasury bills, Treasury bonds, and common stocks in that period? Use the data in Table 11.1. (LO11-2)

Risk, Return, and Capital Budgeting

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

- 12-1 Measure and interpret the market risk, or beta, of a security.
- 12-2 Relate the market risk of a security to the rate of return that investors demand.
- 12-3 Understand why and how project risk determines the opportunity cost of capital.

RELATED WEBSITES FOR THIS CHAPTER CAN BE
FOUND IN CONNECT.



If you want big rewards, you generally need to take big risks.
ZUMA Press, Inc./Alamy Stock Photo

In Chapter 11, we began to come to grips with risk. We stressed the distinction between *specific risk* and *market risk*. Specific risk arises from events that affect only the individual firm or its immediate competitors; it can be eliminated by diversification. But regardless of how much you diversify, you cannot avoid the macroeconomic events that create market risk. This is why investors do not require a higher rate of return to compensate for specific risk but do need a higher return to persuade them to take on market risk.

How can you measure the market risk of a security or a project? We will see why market risk is usually

measured by *beta*—that is, by the sensitivity of the investment's returns to fluctuations in the market. We will also see that the risk premium investors demand should be proportional to beta. This relationship between risk and return is a useful way to estimate the return that investors expect from investing in common stocks.

Finally, we will distinguish the risk of the company's securities and the risk of an individual project. We will also consider what managers should do when the risk of the project is different from that of the company's existing business.

12.1 Measuring Market Risk

market portfolio
Portfolio of all assets in the economy. In practice a broad stock market index is used to represent the market.

beta
Sensitivity of a stock's return to the return on the market portfolio.

Changes in interest rates, government spending, oil prices, foreign exchange rates, and other macroeconomic events affect almost all companies and the returns on almost all stocks. We can, therefore, assess the impact of "macro" news by tracking the rate of return on a **market portfolio** of all securities. If the market is up on a particular day, then the net impact of macroeconomic changes must be positive. We know the performance of the market reflects only macro events because firm-specific events—that is, specific risks—average out when we look at the combined performance of thousands of securities.

In principle, the market portfolio should contain all assets in the world economy—not just stocks but bonds, foreign securities, real estate, and so on. In practice, however, financial analysts make do with indexes of the stock market, such as the Standard & Poor's Composite Index (the S&P 500).¹

Our task here is to define and measure the risk of *individual* common stocks. Because the risk of a diversified portfolio depends on its exposure to macroeconomic, not firm-specific, events, the relevant risk of an individual stock is the degree to which the stock increases the portfolio's exposure to those macro risks. We *measure* the stock's effect on risk as the sensitivity of its returns to fluctuations in the returns of the broad market portfolio, since these fluctuations reflect the net impact of macroeconomic news in any particular period. This sensitivity to fluctuations in the market portfolio is called the stock's **beta**. Beta is often written as the Greek letter β .

Measuring Beta

In the last chapter, we looked at the variability of several individual securities. U.S. Steel had the highest standard deviation, and Coke had the lowest. If you had held U.S. Steel on its own, your returns would have varied about six times as much as they would have if you had held Coke. But wise investors don't put all their eggs in just one basket: They reduce their risk by diversification. An investor with a diversified portfolio will be interested only in the effect each stock has on the risk of the entire portfolio.

Diversification can eliminate the risk that is unique to individual stocks but not the risk that the market as a whole may decline, carrying your stocks with it.

Some stocks are less affected than others by market fluctuations. Investment managers talk about "defensive" and "aggressive" stocks. Defensive stocks are not very sensitive to market fluctuations and, therefore, have low betas. In contrast, aggressive stocks amplify any market movements and have higher betas. If the market goes up, it is good to be in aggressive stocks; if it goes down, it is better to be in defensive stocks (and better still to leave your money in the bank).

Example

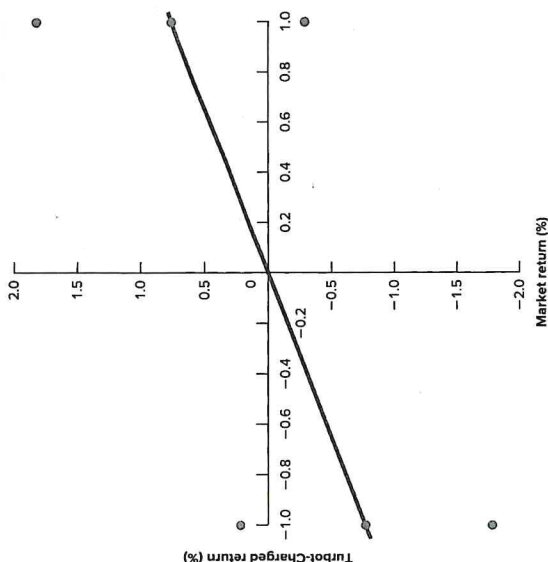
12.1 Measuring Beta for Turbot-Charged Seafoods

Suppose we look back at the trading history of Turbot-Charged Seafoods and pick out six months when the return on the market portfolio was plus or minus 1%.

Month	Market Return (%)	Turbot-Charged Seafoods Return (%)
1	+1	+0.8
2	+1	+1.8
3	+1	-0.2
4	-1	-1.8
5	-1	+0.2
6	-1	-0.8
		Average = 0.8%
		Average = -0.8

¹ We discussed the most popular stock market indexes in Section 11.2.

FIGURE 12.1 This figure is a plot of the data presented in the table in Example 12.1. Each point shows the performance of Turbot-Charged Seafoods stock when the overall market is either up or down by 1%. On average, Turbot-Charged moves in the same direction as the market, but not as far. Therefore, Turbot-Charged's beta is less than 1.0. We can measure beta by the slope of a line fitted to the points in the figure. In this case, it is .8.



Look at Figure 12.1, where these observations are plotted. We've drawn a line through the average performance of Turbot when the market is up or down by 1%. The slope of this line is Turbot's beta. You can see right away that the beta is .8 because, on average, Turbot stock gains or loses .8% when the market is up or down by 1%. Notice that a 2-percentage-point difference in the market return (-1 to +1) generates, on average, a 1.6-percentage-point difference for Turbot shareholders (-.8 to +.8). The ratio, $1.6/2 = .8$, is beta.

In four months, Turbot's returns lie above or below the line in Figure 12.1. The distance from the line shows the response of Turbot's stock returns to news or events that affected Turbot but did not affect the overall market. For example, in month 2, investors in Turbot stock benefited from good macroeconomic news (the market was up 1%) and also from some favorable news specific to Turbot. The market rise gave a boost of .8% to Turbot stock (beta of .8 times the 1% market return). Then, firm-specific news gave Turbot stockholders an extra 1% return, for a total return that month of 1.8%.

Aggressive stocks have high betas, betas greater than 1.0. Their returns tend to amplify returns on the overall market. The betas of defensive stocks are less than 1.0. The returns of these stocks vary less than one-for-one with market returns. The average beta of all stocks is—no surprises here—1.0 exactly.

Now we'll show you how betas are measured.

As Example 12.1 illustrates, we can break down common stock returns into two parts: the part explained by market returns and the firm's beta and the part due to news that is specific to the firm. Fluctuations in the first part reflect market risk; fluctuations in the second part reflect specific risk.

Of course, diversification can get rid of the specific risks. That's why wise investors, who don't put all their eggs in one basket, will look to Turbot's less-than-average beta and call its stock "defensive."

Excel and most other spreadsheet programs provide built-in functions for computing a stock's beta. In columns B and C of the following spreadsheet, we have entered returns for Standard & Poor's 500 Index (the S&P 500) and Amazon for six months in 2020. (In practice, estimates based on just six months would be very unreliable. Estimates of standard deviation and beta use at least 50 returns, usually 60 monthly returns for five years. Sometimes one or two years of weekly returns are used.) This example uses the same data we looked at in the last chapter to estimate the volatility of Amazon. Here, we focus on market risk (beta) rather than total volatility. Let's walk through the spreadsheet.

1. Row 12: Use the SLOPE function to calculate beta. It's important to enter the stock returns (C3:C8) first, followed by the market returns (B3:B8). The beta of 0.66 indicates that during this 6-month period, an extra 1% return on the market was associated with an extra 0.66% return on Amazon.
2. Row 13: The correlation coefficient of .47 reveals that Amazon and the market track each other, but not perfectly. Perfect tracking would imply a correlation coefficient of 1.0. Zero correlation means no tracking at all—in this case, beta would also be zero. The correlation coefficient would

be negative if returns on Amazon and the market tended to move in opposition, one going down when the other is up and one going up when the other is down. However, negative correlations between stocks are rare.

Spreadsheet Questions

1. Suppose that Amazon's return in October 2020 had been 12%, and its return in August 2020 had been -14%. Would you expect its beta to be more or less than the value obtained in the spreadsheet? Reestimate beta with these new data, and confirm your intuition.
2. Suppose that Amazon's return in each month had been 1% higher than the values presented in the accompanying spreadsheet. Would Amazon's beta differ from the value obtained in the spreadsheet? Reestimate beta with these new data, and confirm your intuition.
3. Suppose that you add one more month of data to your original spreadsheet and find that, in that month, Amazon was down 1.6% while the market was up 4.0%. Would you expect Amazon's beta to be more or less than the value obtained in the spreadsheet? What about the correlation coefficient? Reestimate beta and the correlation coefficient with this new data point, and confirm your intuition.

	A	B	C	D
1				
2	Month	S&P 500	Amazon	Formula Used
3	Jul-20	7.01%	10.20%	
4	Aug-20	-3.92%	-10.07%	
5	Sep-20	-2.77%	10.30%	
6	Oct-20	10.75%	8.62%	
7	Nov-20	3.71%	-0.50%	
8	Dec-20	-0.16%	6.97%	
9	Average return	2.44%	4.25%	=AVERAGE(C3:C8)
10	Standard deviation (monthly)	5.26%	7.37%	=STDEV.P(C3:C8)
11	Standard deviation, annualized	18.22%	25.54%	=C10*SQRT(12)
12	Beta		0.66	=SLOPE(C3:C8,B3:B8)
13	Correlation		0.47	=CORREL(B3:B8,C3:C8)

BEYOND THE PAGE



Calculating beta

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12.1 Self-Test

Here are six months' returns to stockholders in the Anchovy Queen restaurant chain:

Month	Market Return (%)	Anchovy Queen Return (%)
1	+1	+2.0
2	+1	+0
3	+1	+1.0
4	-1	-1.0
5	-1	+0
6	-1	-2.0

Draw a figure like Figure 12.1 and check the slope of the fitted line. What is Anchovy Queen's beta?

BEYOND THE PAGE



Calculating Beta

www.mhhe.com/brealey11e

BEYOND THE PAGE



Comparing beta estimates

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Real life doesn't serve up numbers quite as convenient as those in our examples so far. However, the procedure for measuring real companies' betas is exactly the same:

1. Observe rates of return, usually monthly or weekly, for the stock and the market.
2. Plot the observations as in Figure 12.1.
3. Fit a line showing the average return to the stock at different market returns.

Beta is the slope of the fitted line.

This may sound like a lot of work, but in practice, computers do it for you. The nearby box shows how to use the SLOPE function in Excel to calculate a beta. Here are two real examples.

Betas for Amazon and McDonald's

Each point in Figure 12.2a shows the return on Amazon stock and the return on the market index in a different month. For example, the upper highlighted point shows that in October 2017, Amazon's stock price rose by 15%, whereas the market index rose by 2.2%. Notice that more often than not, Amazon outperformed the market when the index rose and underperformed the market when the index fell. Thus, Amazon was a relatively aggressive, high-beta stock.

We have drawn a line of best fit through the points in the figure.² The slope of this line is 1.51. For each extra 1% rise in the market, Amazon's stock price moved on average an extra 1.51%. For each extra 1% fall in the market, Amazon's stock price fell an extra 1.51%. Thus, Amazon's beta was 1.51.³

Of course, Amazon's stock returns are not perfectly related to market returns. For example, in October 2017, the point highlighted in Figure 12.2a, Amazon performed better than one would have predicted given the return on the market index. We know this because Amazon's return in that month lies above the upward-sloping line in the figure—the line that describes the typical relation between the market return and Amazon's return. The vertical distance from this "beta line" to Amazon's return in October 2017 shows the impact of the firm-specific events that lifted Amazon's fortunes in that month. At other times, such as November 2016, Amazon flew south when the market went north. In that month, firm-specific events conspired to reduce Amazon's return.

Thus, while the slope of the line in Figure 12.2a measures beta and exposure to market risk, firm-specific risk shows up in the scatter of points around the line: Wider scatter means more firm-specific risk.

Figure 12.2b shows a similar plot of the monthly returns for McDonald's. In contrast to Amazon, McDonald's was a defensive, low-beta stock. It was not highly sensitive to market movements, usually lagging when the market rose, but doing better (or less badly) when the market fell. The slope of the line of best fit shows that on average an extra 1% change in the index resulted in an extra .45% change in the price of McDonald's stock. Thus, McDonald's had a beta of .45.

Estimates of beta can be accessed easily, for example, at finance.yahoo.com, but you may find it interesting to look at Table 12.1, which shows betas of several well-known stocks. Newmont Mining had the lowest beta: Its stock price actually had a (very slight) tendency to fall when the market rose. U.S. Steel was at the other extreme: Its return was 3.03 times as sensitive as the average stock to market movements.

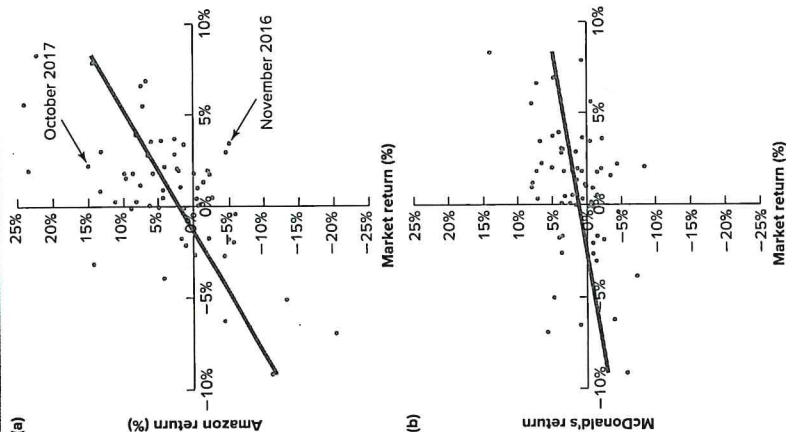
Total Risk and Market Risk

U.S. Steel tops our list of betas in Table 12.1. It was also at the top of Table 11.5, which showed the total variability of the same group of stocks. But total risk is not the

² The line of best fit is a regression line. The slope of the line can be calculated using *ordinary least squares* regression. The dependent variable is the return on the stock (Amazon). The independent variable is the return on the market index. In this case, the S&P 500.

³ This estimate of Amazon's beta differs from the estimate in the Spreadsheet Solutions box. We warned you there that estimates based on six months of data can be very unreliable.

FIGURE 12.2 (a) Each point in this figure shows the returns on Amazon common stock and the overall market in a particular month between January 2015 and December 2019. Amazon's beta is the slope of the line fitted to these points. Amazon has an above-average beta of 1.51. (b) In this plot of 60 months' returns for McDonald's and the overall market, the slope of the fitted line is much less than Amazon's beta in panel a. McDonald's has a low beta of .45.



same as market risk. Some of the most variable stocks have below-average betas, and vice versa.

Consider, for example, Newmont Mining. Newmont is the world's largest gold producer. Its annual report cites many risks the company faces, including, for example, price fluctuations in the price of gold and other metals, currency fluctuations, uncertainty in the cost of its operations, and political risk and changes in tax laws in the countries where it operates.

These risks are considerable and are reflected in the high standard deviation of the returns on Newmont's stock (see Table 11.5). But they are mostly firm-specific, not macro, risks. When the U.S. economy is booming, gold prices are just as likely to slump. So, while Newmont stock has above-average total volatility, it has a relatively low beta.

Firm-specific risk is, of course, diversifiable and of no concern to an investor tracking the performance of his or her highly diversified portfolio. Newmont's CEO and financial managers are acutely interested in firm-specific risk, however, always hoping

TABLE 12.1 Betas for selected common stocks, January 2015–December 2019.

Ticker	Company	Beta
X	U.S. Steel	3.03
MRO	Marathon Oil	2.35
AMZN	Amazon	1.51
IBM	IBM	1.33
BA	Boeing	1.19
F	Ford	1.09
UNP	Union Pacific	1.07
GOOG	Google/Alphabet	1.01
DIS	Disney	1.00
XOM	ExxonMobil	1.00
INTC	Intel	0.91
PFE	Pfizer	0.65
PCG	Pacific Gas & Electric	0.55
SBUX	Starbucks	0.51
MCD	McDonald's	0.45
CPB	Campbell Soup	0.43
KO	Coca-Cola	0.43
WMT	Walmart	0.35
NEM	Newmont Mining	-0.01
Average		0.99

Note: Betas are calculated from five years of monthly data.

that the next return will plot above the beta line. Investors analyzing Newmont's financial performance will likewise watch total return, even if much of the risk in that return will, in the end, be diversified away.

12.2

What Can You Learn from Beta?

You can learn a lot from beta. First, if you don't know whether a stock is defensive or aggressive, you can check whether its beta has been less than or greater than 1.0. Second, you can predict the beta of a portfolio.

Portfolio Betas

The beta of a portfolio is just an average of the betas of the individual securities in the portfolio, weighted by the investment in each security. For example, the beta of a two-stock portfolio is calculated as:

$$\text{Portfolio beta} = (\text{fraction of portfolio in stock 1} \times \text{beta of stock 1}) + (\text{fraction of portfolio in stock 2} \times \text{beta of stock 2}) \quad (12.1)$$

Thus, a portfolio invested 50–50 in Amazon and McDonald's would have a beta of $(.5 \times 1.51) + (.5 \times .45) = .98$.

Suppose you formed a portfolio of the stocks in Table 12.1 with an equal amount invested in each stock. You could predict the portfolio beta as a simple average of the betas listed in the table. If you decided to invest more money in some stocks than others, you would have to calculate a weighted average.

12.2 Self-Test

Rosa Rugonis receives a bequest of \$1 million and decides to invest it as follows: \$200,000 each in Ford, Starbucks, Union Pacific, and IBM, and \$100,000 each in Newmont Mining and Walmart. Rosa is, of course, undiversified. But what is her portfolio beta? Use the stock betas in Table 12.1.

Example 12.2

How Risky Are Mutual Funds?

You don't have to be wealthy to own a diversified portfolio. You can buy shares in one of the more than 8,000 mutual funds in the United States.

Investors buy shares of the funds, and the funds use the money to buy portfolios of securities. The returns on the portfolios are passed through to the funds' owners in proportion to their shareholdings. Therefore, the funds act like investment cooperatives, providing even the smallest investors with diversification and professional management at low cost.

Panel a of Figure 12.3 shows the calculation of beta for both an individual stock (Union Pacific railroad), and a stock mutual fund, Vanguard's Growth & Income (G&I) Fund. As in Figure 12.1, each dot in the diagram shows the return in a particular month for the S&P 500 and a particular investment, either Union Pacific or the mutual fund. The orange dots are for Union Pacific, and the blue dots are for the mutual fund. During this period, both Union Pacific and the Vanguard fund displayed approximately average sensitivity to market fluctuations, with betas of 1.07, only slightly above 1. Therefore, if Union Pacific and the G&I fund had no specific risk, their returns would have been about as variable as the market's. But they both do have specific risk, so their returns actually were more variable than the market's.

FIGURE 12.3 (a) Betas of Union Pacific and Vanguard's Growth & Income Fund. Both the fund and the stock have betas of 1.07, but the fund has far less specific risk, reflected in the tighter cluster of the black dots around the line.

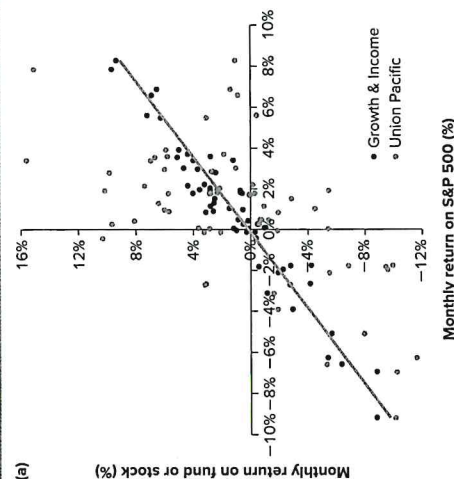
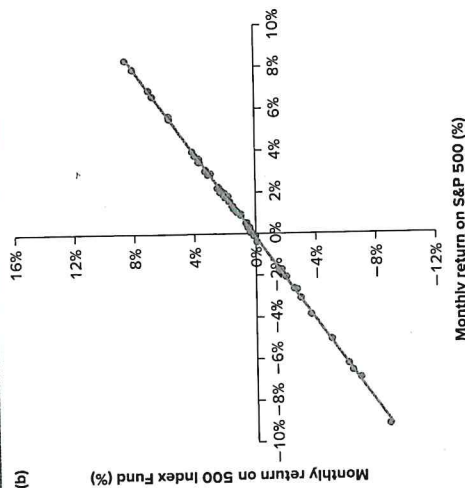


FIGURE 12.3 (b) The beta of Vanguard's 500 Index Fund. This is a fully diversified index fund designed to track the performance of the market. Note the index fund's beta (1.0) and the near absence of specific risk. The fund's returns lie almost precisely on the fitted line relating its returns to those of the S&P 500 portfolio.



Union Pacific's returns exhibit a wide scatter, reflecting its considerable firm-specific risk. As a result, its total volatility is substantially greater than that of the market. Turn back to Table 11.5 and you will see that its standard deviation was 21.4% annually, versus only 12.0% for the S&P 500. Vanguard's G&I fund is highly diversified—it holds more than 800 different stocks—and its dots cluster tightly around the fitted line. Its total volatility, 13.4%, is not much more than that of the market and is considerably less than Union Pacific's; this is the benefit of diversification.

How did the G&I fund achieve a beta of 1.07? By acquiring stocks with an average beta of 1.07. Remember, a portfolio beta equals the weighted-average beta of its component securities. The weights are the proportions invested in each security. If G&I were not a mutual fund (whose returns we can directly observe and plot as in Figure 12.3) but instead was a privately held portfolio—say, a private pension fund—we could calculate its beta as the weighted-average beta of its holdings.

Figure 12.3b, shows a similar plot for Vanguard's 500 Index mutual fund. This fund has a beta of 1.0 and only the tiniest level of specific risk—each month's return on the fund matches almost exactly that of the index. The managers of an index fund do not attempt to pick stocks but just work to achieve full diversification at very low cost. Investors in this fund buy the market as a whole and don't need to worry at all about specific risk. ■

By the way, knowing a mutual or pension fund's beta will help you ask the right questions when a portfolio manager brags about "beating the market." Suppose, for example, that you met the manager of a mutual fund that delivered a 25% rate of return in 2020, four percentage points above the 21% return on the broad market index in that year. Should you have congratulated the fund manager for her stock-picking prowess? It depends on the fund's beta. Suppose the fund specialized in high-beta stocks with an average beta of 1.5. Then you should *not* have been impressed. The rise in the market alone should have contributed $1.5 \times 21\% = 31.5\%$ to the return on the mutual fund. So

the manager's return of 25% was actually disappointing. The mutual fund's shareholders would have been better off in a high-beta index fund.

Of course, you cannot judge a mutual fund manager's skill based on only one year's performance. We simply point out that "beating the market" is child's play for a high-beta portfolio in a bull market like 2020.

The Portfolio Beta Determines the Risk of a Diversified Portfolio

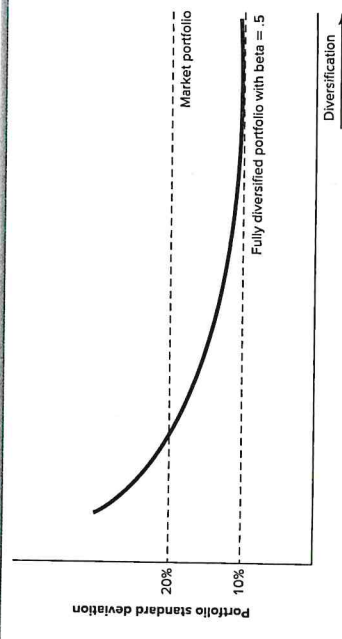
You can also use beta to predict the *total risk* (standard deviation) of a diversified portfolio. Look back to Figure 11.9, which shows how the volatility of a portfolio return falls as more stocks are added and the portfolio becomes better diversified. The gain from diversification is rapid at first but slows down as specific risks are diversified away. With full diversification, only market risk remains.

Okay, but how much market risk remains? It depends on the beta of the portfolio. Suppose you construct a diversified portfolio of stocks with betas clustered around .5. The portfolio beta will also be .5. Most of the individual stocks' specific risk would be diversified away. The market risk would remain, and the portfolio would end up half as variable as the market. If the standard deviation of the market is 20%, a fully diversified portfolio of stocks with betas of .5 has a standard deviation of $.5 \times 20 = 10\%$. We show this outcome in Figure 12.4. Suppose, on the other hand, you construct a fully diversified portfolio of stocks with betas clustered around 1.3. In that case the portfolio standard deviation would be $1.3 \times 20 = 26\%$.⁴

Of course, the average beta across all stocks must be 1.0. A fully diversified portfolio, including all stocks, will match the market and so have a beta of 1.0 and the same standard deviation as the market.

A wise and diversified investor does not judge the risk of individual stocks by their stand-alone volatility, but by their contributions to *portfolio* risk. By now you can see

FIGURE 12.4 The risk of a fully diversified portfolio depends on the portfolio beta. In this example, the portfolio beta is .5 and the standard deviation of the portfolio "bottoms out" at 10%, half the market standard deviation of 20%.



⁴ It can be difficult to achieve full diversification when constructing portfolios using stocks with very high or low betas. For example, a portfolio constructed solely from stocks with very high betas, such as U.S. Steel, could be concentrated in cyclical industries, and thus exposed to some undiversified industry risk in addition to market risk.

that the contributions depend on the stocks' betas. In the next sections, we will see how that observation leads to a model of risk and return and a way to estimate the opportunity cost of capital.

12.3 Self-Test

Suppose you construct a portfolio from only 10 stocks with an average beta of 1.2. The market standard deviation is 20%. What can you say about the portfolio's standard deviation? Is it 24%? Less? More? Why?

12.3

Risk and Return

In Chapter 11, we looked at past returns on selected investments. The least risky investment was U.S. Treasury bills. Because the return on Treasury bills is fixed, it is unaffected by what happens to the market. Thus, the beta of Treasury bills is zero. The *riskiest* investment that we considered was the market portfolio of common stocks. This has average market risk: Its beta is 1.0.

Wise investors don't run risks just for fun. They are playing with real money and, therefore, require a higher average return from the market portfolio than from Treasury bills. The difference between the return on the market and the interest rate on bills is termed the **market risk premium**. Since 1900, the average market risk premium has been 7.8% a year. Of course, there is plenty of scope for argument as to whether the past century constitutes a typical period, but we will just assume here that the normal risk premium is a nice round 7%; that is, 7% is the additional return that an investor could reasonably expect on average from investing in the stock market rather than in Treasury bills.

In Figure 12.5a, we have plotted the risk and expected return from Treasury bills and the market portfolio. You can see that Treasury bills have a beta of zero and a risk-free return; we'll assume that return is 1%. The market portfolio has a beta of 1.0, an expected risk premium of 7%, and an expected return of $1 + 7 = 8\%$.

Now, given these two benchmarks, what expected rate of return should an investor require from a portfolio that is equally divided between Treasury bills and the market? Halfway between, of course. Thus, in Figure 12.5b, we have drawn a straight line through the Treasury bill return and the expected market return. The portfolio (marked with an X) would have a beta of .5 and an expected return of 4.5%. This includes a risk premium of 3.5% above the Treasury bill return of 1%.

You can calculate this return as follows: Start with the difference between the expected market return r_m and the Treasury bill rate r_f . This is the expected market risk premium:

$$\text{Market risk premium} = r_m - r_f = 8\% - 1\% = 7\%$$

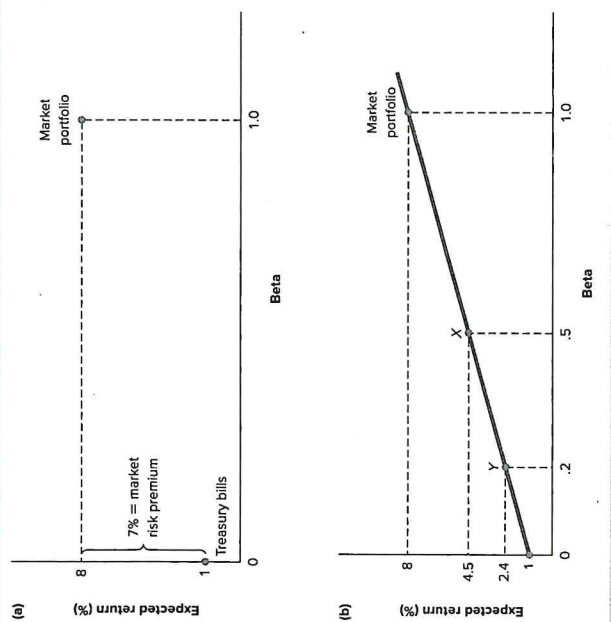
Beta measures risk relative to the market. Therefore, the portfolio's expected risk premium equals its beta times the market risk premium:

$$\text{Risk premium} = r - r_f = \beta(r_m - r_f)$$

For example, with a beta of .5 and a market risk premium of 7%,

$$\text{Risk premium} = \beta(r_m - r_f) = .5 \times 7\% = 3.5\%$$

FIGURE 12.5 (a) Here we begin the plot of expected rate of return against beta. The first benchmarks are Treasury bills (beta = 0) and the market portfolio (beta = 1.0). We assume a Treasury bill rate of 1% and a market return of 8%. The market risk premium is $8 - 1 = 7\%$. (b) A portfolio split evenly between Treasury bills and the market will have beta = .5 and an expected return of 4.5% (point X). A portfolio invested 20% in the market and 80% in Treasury bills has beta = .2 and an expected rate of return of 2.4% (point Y). Note that the expected rate of return on any portfolio mixing Treasury bills and the market lies on a straight line. The risk premium is proportional to the portfolio beta.



The total expected rate of return is the sum of the risk-free rate and the risk premium:

$$\begin{aligned} \text{Expected return} &= \text{risk-free rate} + \text{expected risk premium} \\ r &= r_f + \beta(r_m - r_f) \\ &= 1\% + 3.5\% = 4.5\% \end{aligned} \quad (12.2)$$

You could have calculated the expected rate of return in one step from this formula:

$$\begin{aligned} \text{Expected return} &= r = r_f + \beta(r_m - r_f) \\ &= 1\% + (.5 \times 7\%) = 4.5\% \end{aligned}$$

This basic relationship should hold not only for our portfolios of Treasury bills and the market, but for any asset. This conclusion is known as the **capital asset pricing model**, or **CAPM** (pronounced CAP-EM). The CAPM has a simple interpretation: The expected rates of return demanded by investors depend on two things: (1) compensation for the time value of money (the risk-free rate, r_f) and (2) a risk premium, which depends on beta and the market risk premium.

Note that the expected rate of return on an asset with $\beta = 1.0$ is just the market return. With a risk-free rate of 1% and market risk premium of 7%,

$$\begin{aligned} r &= r_f + \beta(r_m - r_f) \\ &= 1\% + (1 \times 7\%) = 8\% \end{aligned}$$

capital asset pricing model (CAPM)

Theory of the relationship between risk and return stating that the expected risk premium on any security equals its beta times the market risk premium.

12.4 Self-Test

What are the risk premium and expected rate of return on a stock with $\beta = 1.5$? Assume a Treasury bill rate of 1% and a market risk premium of 7%.

Why the CAPM Makes Sense

The CAPM assumes that the stock market is composed of well-diversified investors—investors operating at the bottom of the portfolio risk curves illustrated in Figures 11.9 and 12.4. The risks borne by such investors depend on their portfolio betas. This is realistic in a market dominated by large institutions, where even small fry can use mutual or exchange-traded funds to diversify at very low cost. The following example shows why this view of risk leads to the CAPM.

Example 12.3 How Would You Invest \$1 Million?

Have you ever daydreamed about receiving a \$1 million check, no strings attached, from an unknown benefactor? Let's daydream about how you would invest it.

We have two good candidates: Treasury bills, which offer an absolutely safe return, and the market portfolio (possibly via an index fund). The market has generated higher returns on average, but those returns have fluctuated a lot. (Look back to Figure 11.4.) So, your investment policy is going to depend on your tolerance for risk.

If you're a wimp, you may invest only a small part of your money in the market portfolio and lend the remainder to the government by buying Treasury bills. Suppose that you invest 20% of your money in the market portfolio and put the other 80% in U.S. Treasury bills. Then the beta of your portfolio will be a mixture of the beta of the market ($\beta_{\text{market}} = 1.0$) and the beta of the T-bills ($\beta_{\text{T-bills}} = 0$):

$$\begin{aligned} \text{Beta of portfolio} &= \left(\frac{\text{proportion}}{\text{in market}} \times \text{beta of market} \right) + \left(\frac{\text{proportion}}{\text{in T-bills}} \times \text{beta of T-bills} \right) \\ \beta &= (.2 \times \beta_{\text{market}}) + (.8 \times \beta_{\text{T-bills}}) \\ &= (.2 \times 1.0) + (.8 \times 0) \\ &= .20 \end{aligned}$$

The fraction of funds that you invest in the market also affects your expected return. If you invest your entire million in the market portfolio, you earn the full market risk premium. But if you invest only 20% of your money in the market, your portfolio beta is .20 and you earn only 20% of the market risk premium.

$$\begin{aligned} \text{Expected risk premium on portfolio} &= \left(\frac{\text{proportion}}{\text{in market}} \times \text{market risk premium} \right) + \left(\frac{\text{proportion}}{\text{in T-bills}} \times \text{risk premium on T-bills} \right) \\ &= (.2 \times \text{expected market risk premium}) + (.8 \times 0) \\ &= .2 \times \text{expected market risk premium} \\ &= .2 \times 7 = 1.4\% \end{aligned}$$

Notice that the portfolio risk premium equals the product of beta times the market risk premium.

The expected return on your portfolio is equal to the risk-free interest rate plus the expected risk premium:

$$\text{Expected portfolio return} = r_{\text{portfolio}} = 1 + 1.4 = 2.4\%$$

The expected return on this passive portfolio, therefore, gives a benchmark for the fair expected return on any other investment or asset with the same risk (beta). In Figure 12.5, panel b, we show the beta and expected return on this portfolio by the letter Y. ■

12.5 Self-Test

How would you construct a diversified portfolio with a beta of .25? What is the expected return to this strategy? Assume Treasury bills yield 1% and the market risk premium is 7%.

The Security Market Line

Example 12.3 illustrates a general point: By investing some proportion of your money in the market portfolio and lending (or borrowing)³ the balance, you can obtain any combination of risk and expected return along the sloping line in Figure 12.6. This line is known as the security market line.

The security market line describes the expected returns and risks from splitting your overall portfolio between risk-free securities and the market. It also sets a standard for other investments. Investors will be willing to hold other securities only if they offer equally good prospects. Thus, the required risk premium for any investment is given by the security market line:

$$\text{Risk premium on investment} = \text{beta} \times \text{expected market risk premium}$$

Look back to Figure 12.5b, which suggests that an individual common stock with $\beta = .5$ must offer a 4.5% expected rate of return when Treasury bills yield 1% and the market risk premium is 7%. You can now see why this has to be so. If that stock offered a lower rate of return, nobody would buy even a little of it—they could get 4.5%

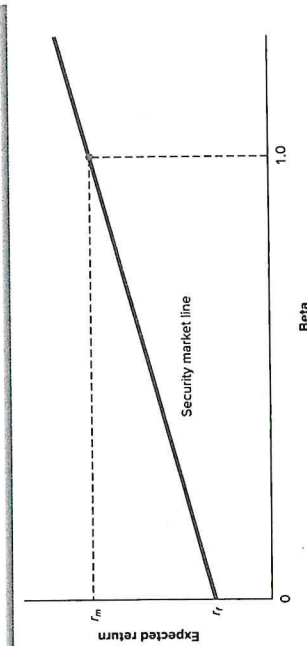


FIGURE 12.6 The security market line shows how expected rate of return depends on beta. According to the capital asset pricing model, expected rates of return for all securities and all portfolios lie on this line.

12.6 Self-Test

Suppose you invest \$400,000 in Treasury bills and \$600,000 in the market portfolio. What is the return on your portfolio if bills yield 3% and the expected return on the market is 10%? What does the return on this portfolio imply for the expected return on individual stocks with betas of .6?

Using the CAPM to Estimate Expected Returns

We can use the CAPM to calculate expected rates of return that investors demand on individual stocks or other securities. We need three numbers: the risk-free interest rate, the expected market risk premium, and beta. Suppose the interest rate on Treasury bills is 1% and the expected market risk premium is 7%. We take betas from Table 12.1. Table 12.2 puts these numbers together in the CAPM to get expected rates of return. Take Campbell Soup as an example:

$$\begin{aligned} \text{Expected return on Campbell} &= \text{risk-free interest rate} + \text{beta} \times \text{market risk premium} \\ r &= r_f + \beta \times (r_m - r_f) = 1\% + .43 \times 7\% = 4.0\% \end{aligned}$$

U.S. Steel had the highest beta in our sample of companies and the highest required return of about 22.2%. At the other extreme, Newmont Mining had a very low beta and a low CAPM expected rate of return.

The calculations in Table 12.2 are easy, but don't assume that using the CAPM is a purely mechanical exercise. First, betas are estimates, not exact measurements. Campbell's beta of .43 is a best statistical estimate, but its true beta could well be quite a bit higher or lower. If you want to round it to .4, that's fine with us. Second, very high or very low betas tend not to repeat in the future. Remember that beta measures how sensitive a particular firm is to the macroeconomy and that sensitivity can change as the risks facing the economy change. For example, airline betas surged as both leisure and business travel slumped during the Covid crisis of 2020. The nearby box shows how many other betas changed dramatically—some increasing and others decreasing—as Covid realigned the positions of firms in the broad economy. Third, it's difficult to pin down the expected future market risk premium. Fourth, while the CAPM is widely used in practice, it is not the last word in risk and return in the stock market, as we will now see.

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COVID-19 and Risk

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³ Notice that the security market line extends above the market return at $\beta = 1.0$. How would you generate a portfolio with, say, $\beta = 2.0$? It's easy, but it's risky. Suppose you borrow \$1 million and invest the loan plus \$1 million in the market portfolio. That gives you \$2 million invested and a \$1 million liability. Your portfolio now has a beta of 2.0.

$$\begin{aligned} \text{Beta of portfolio} &= (\text{proportion in market} \times \text{beta of market}) + (\text{proportion in loan} \times \text{beta of loan}) \\ \beta &= (2 \times \beta_{\text{market}}) + (-1 \times \beta_{\text{loan}}) \\ &= (2 \times 1.0) + (-1 \times 0) = 2 \end{aligned}$$

Notice that the proportion in the loan is negative because you are borrowing, not lending, money. By the way, borrowing from a bank or stockbroker would not be difficult or unduly expensive as long as you put up your \$2 million stock portfolio as security for the loan. Can you calculate the risk premium and the expected rate of return on this borrow-and-invest strategy?

Beta tells us how the prospects of a particular firm respond to those of the broad economy. But don't think that betas are carved in stone. As the economy changes, so will beta.

Think about Zoom Video Communications. In 2019, Zoom catered primarily to corporate clients, which used its services for videoconferencing and telecommuting. Zoom was a typical cyclical firm, with a business model tied to the general level of business activity; not surprisingly, its beta was above 1, specifically 1.82.⁶

Then came the coronavirus and the "work from home" economy. Suddenly, Zoom, which few people had heard of before the pandemic, became an essential tool for both work and socializing. The worse the prognosis for the pandemic (and, as a result, for the economy), the greater was the demand for Zoom's services. Zoom suddenly became a

countercyclical firm, and its beta estimated between February 20 and June 1 was actually negative, -0.30 .

Zoom was not alone. The biotech firm Regeneron Pharma saw its beta fall from above 1 to below -0.5 . Again, the worse the prospects for the pandemic and the economy, the more highly investors seemed to value Regeneron's specialized expertise.

At the other extreme, some betas skyrocketed. While the economy was struggling, Carnival Cruise Line faced disaster. Its prospects were painfully tied to those of the economy and the end of the pandemic, and its beta nearly doubled, from a bit above 1 to nearly 2.

The lesson? You can estimate past beta from historical returns, but you also need to apply some judgment. Don't assume that the past necessarily will be prologue.

TABLE 12.2 Expected rates of return demanded by investors for selected companies. Expected return is calculated assuming a risk-free rate of 1% and a market risk premium of 7%.

Ticker	Company	Beta	Expected Return (%)
X	U.S. Steel	3.03	22.2
MRO	Marathon Oil	2.35	17.4
AMZN	Amazon	1.51	11.5
IBM	IBM	1.33	10.3
BA	Boeing	1.19	9.3
F	Ford	1.09	8.6
UNP	Union Pacific	1.07	8.5
GOOG	Google/Alphabet	1.01	8.1
DIS	Disney	1.00	8.0
XOM	ExxonMobil	1.00	8.0
INTC	Intel	0.91	7.3
PFE	Pfizer	0.65	5.6
PCG	Pacific Gas & Electric	0.55	4.9
SBUX	Starbucks	0.51	4.5
MCD	McDonald's	0.45	4.1
CPB	Campbell Soup	0.43	4.0
KO	Coca-Cola	0.43	4.0
WMT	Walmart	0.35	3.5
NEM	Newmont Mining	-0.01	0.9
Average		0.99	7.94

Note: Expected return = $1\% + \text{beta} \times 7\%$.

How Well Does the CAPM Work?

The basic idea behind the capital asset pricing model is that investors expect a reward for both waiting and worrying. The greater the worry, the greater the expected return. If you invest in a risk-free Treasury bill, you just receive the rate of interest. That's the reward for waiting. When you invest in risky stocks, you demand an extra return or risk premium for worrying. When risk is higher, investors will not be willing to pay as much for the shares of the stock, and the price will fall until the rate of return they can expect to earn provides a "fair" expected risk premium. The capital asset pricing model states that this risk premium is equal to the stock's beta times the market risk premium.

⁶ These estimates of beta were calculated by Derek Horstmeier and Chaitanya M. Vij and reported in Karen Langley, "For Some Stocks, Bad News Is Now Good News," *The Wall Street Journal*, July 20, 2020.

How well does the CAPM work in practice? For example, is the expected return on stocks with betas of $.5$ halfway between the return on the market portfolio and the interest rate on Treasury bills? Unfortunately, the evidence is conflicting. Let's look back to the actual returns earned by investors in low-beta stocks and in high-beta stocks.

Imagine that 10 investors gathered together in a Wall Street bar in 1931 and agreed to establish investment trust funds for their children. Each investor decided to follow a different strategy. Investor 1 opted to buy the 10% of the New York Stock Exchange stocks with the lowest estimated betas; investor 2 chose the 10% with the next-lowest betas; and so on, up to investor 10, who proposed to buy the stocks with the highest betas. They also planned that at the end of each year they would reestimate the betas of all NYSE stocks and reconstitute their portfolios. And so they parted with much cordiality and good wishes.

In time, the 10 investors all passed away, but their children agreed to meet in 2020 in the same bar to compare the performance of their portfolios. Figure 12.7 shows how they fared. Investor 1's portfolio turned out to be much less risky than the market; its beta was only $.48$. However, investor 1 also realized the lowest return, 8.1% above the risk-free rate of interest. At the other extreme, the beta of investor 10's portfolio was 1.55 , about three times that of investor 1's portfolio. But investor 10 was rewarded with the highest return, averaging 14.6% a year above the interest rate. So, over this 89-year period, returns did indeed increase with beta.

As you can see from Figure 12.7, the market portfolio over the same 89-year period provided an average return of 11.9% above the interest rate⁷ and (of course) had a beta of 1.0 . The CAPM predicts that the risk premium should increase in proportion to beta, so the returns of each portfolio should lie on the upward-sloping security market line in Figure 12.7. Because the market provided a risk premium of 11.9% , investor 1's portfolio, with a beta of $.48$, should have provided a risk premium of 5.7% and investor 10's portfolio, with a beta of 1.55 , should have given a premium of 18.5% . You can see that, while high-beta stocks performed better than low-beta stocks, the difference was not as great as the CAPM predicts.

Figure 12.7 provides broad support for the CAPM, though it suggests that the line relating return to beta has been too flat. But recent years have been less kind to the CAPM. For example, if the 10 friends had invested their cash in 1966 rather than

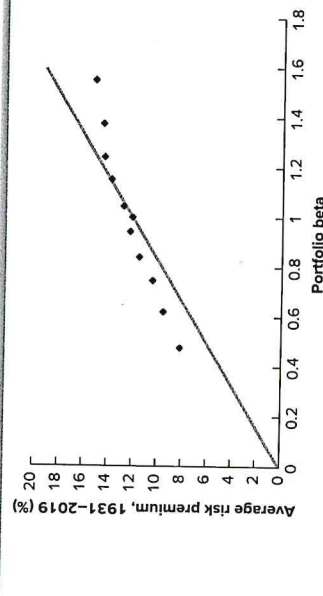


FIGURE 12.7 The capital asset pricing model states that the expected risk premium from any investment should lie on the security market line. The dots show the actual average risk premium from portfolios with different betas. The high-beta portfolios generated higher returns, just as predicted by the CAPM. But the high-beta portfolios plotted below the security market line and the low-beta portfolios plotted above. A line fitted to the 10 portfolio returns would be "flatter" than the security market line.

Source: This is an update of calculations that originally appeared in F. Black, "Beta and Return," *Journal of Portfolio Management* 20 (Fall 1993), pp. 8-18. We are grateful to Adam Kolinski for recalculating and extending the plots.

⁷ In Figure 12.7, the stocks in the "market portfolio" are weighted equally. Because the stocks of small firms have provided higher average returns than those of large firms, the risk premium on an equally weighted index is higher than that on a value-weighted index. This is one reason for the difference between the 11.9% market risk premium in Figure 12.7 and the 7.8% premium reported in Table 11.1.

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The momentum factor

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1931, there would have been very little relation between their portfolio returns and beta.⁸ Does this imply that there has been a fundamental change in the relation between risk and return in the last 51 years, or did high-beta stocks just happen to perform worse during these years than investors initially expected? It is hard to be sure.

There is little doubt that the CAPM is too simple to capture everything that is going on in the market. For example, look at Figure 12.8. The orange line shows the cumulative difference between the returns on small-firm stocks and large-firm stocks. If you had bought the shares with the smallest market capitalizations and sold those with the largest capitalizations, this is how your wealth would have changed. You can see that small-cap stocks did not always do well, but over the long haul, their owners have made substantially higher returns. Since the end of 1926, the average annual difference between the returns on the two groups of stocks has been 2.4%. Now look at the blue line in Figure 12.8, which shows the cumulative difference between the returns on value stocks and growth stocks. *Value stocks* here are defined as those with high ratios of book value to market value. *Growth stocks* are those with low ratios of book to market. Notice that value stocks have provided a higher long-run return than growth stocks. Since 1926, the average annual difference between returns on value and growth stocks has been about 3.9%.

The superior performance of small-firm stocks and value stocks does not fit well with the CAPM, which predicts that beta is the only reason that expected returns differ. If investors expected the returns to depend on firm size or book-to-market ratios, then the simple version of the capital asset pricing model cannot be the whole truth.

What's going on here? It is hard to say. Defenders of the capital asset pricing model emphasize that it is concerned with *expected* returns, whereas we can observe only *actual* returns. Actual returns reflect expectations, but they also embody lots of "noise"—the steady flow of surprises that conceal whether on average investors have received the returns that they expected. Thus, when we observe that in the past small-firm stocks and value stocks have provided superior performance, we can't be sure whether this was simply a coincidence or whether investors have required a higher expected return to hold these stocks.

Such debates have prompted headlines like "Is Beta Dead?" in the business press. It is not the first time that beta has been declared dead, but the CAPM remains the leading model for estimating required returns. Only strong theories can have more than one funeral.

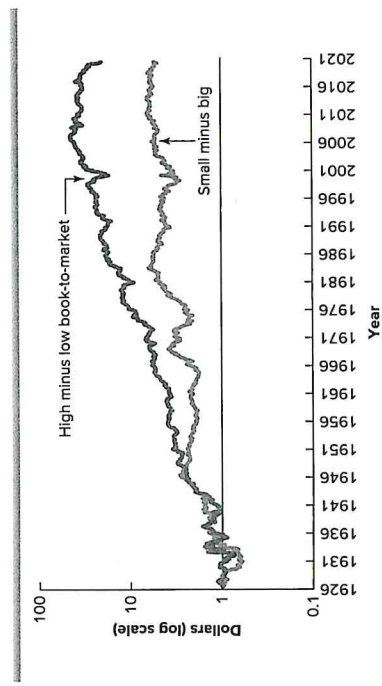


FIGURE 12.8 The orange line shows the cumulative difference between the returns on small-firm and large-firm stocks from 1926 to 2020. The blue line shows the cumulative difference between the returns on high-book-to-market-value stocks and low-book-to-market-value stocks.

Source: mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

⁸ During this later period, the returns to the first seven investors increased in line with beta. However, the highest-beta portfolios performed poorly.

The CAPM is not the only model of risk and return. It has several brothers and sisters as well as second cousins. However, the CAPM captures in a simple way two fundamental ideas. First, almost everyone agrees that investors require some extra return for taking on risk. Second, investors appear to be concerned principally with the market risk that they cannot eliminate by diversification. That is why nearly three-quarters of financial managers use the capital asset pricing model to estimate the cost of capital.⁹

The CAPM and the Opportunity Cost of Capital

The discount rate for valuing a proposed capital investment project should be the *opportunity cost of capital*, defined as the expected rate of return that the company's shareholders could achieve by investing on their own. But the CAPM tells us (confirming common sense) that expected rates of return depend on risk, that is, on beta. Therefore, the opportunity cost of capital for a proposed project should depend on the project's beta. The **project cost of capital** is, therefore, its minimum acceptable expected rate of return, given its risk.

12.4

project cost of capital
Minimum acceptable
expected rate of return on
a project given its risk.

Example 12.4

Estimating the Opportunity Cost of Capital for a Project

Suppose that Coca-Cola is contemplating an investment of \$10 million to expand a warehouse. It has forecast cash flows and calculated an internal rate of return (IRR) of 3%. We assume, as in Table 12.2, that the risk-free rate is 1% and the market risk premium is 7%. Should Coke invest?

To answer this question, you need to know the opportunity cost of capital r , defined as the expected rate of return that Coke's shareholders could achieve if the \$10 million were paid out to them to invest on their own. The CAPM says that the expected rate of return demanded by investors depends on the project beta.

Suppose that the project's cash flows are an absolutely sure thing. Then, beta = 0 and the project cost of capital is

$$r = r_f + \beta \times (r_m - r_f) = 1\% + 0 \times 7\% = 1\%$$

If the project offers a 3% return when the opportunity cost of capital for the project is 1%, Coke should obviously go ahead.¹⁰

What about the 4.0% rate of return for Coke in Table 12.2? It's irrelevant if we are confident that the project is risk-free. Coke's stock offers more than the risk-free rate because it is risky, with a beta of .43. Coke's stockholders would all vote for Coke to invest on their behalf in a risk-free project earning 3% because the stockholders could get only 1% investing risk-free on their own.

Surefire projects rarely occur outside finance textbooks. So let's think about the opportunity cost of capital if the project has the same risk as the market portfolio. In this case, beta is 1.0, and the cost of capital is the expected rate of return on the market.

$$r = r_f + \beta \times (r_m - r_f) = 1\% + 1.0 \times 7\% = 8\%$$

Now that the project beta is 1.0, the project is no longer worth doing. The stockholders would vote against the project at a 3% expected return because they could earn a higher return, 8%, at the same level of risk (same beta) by investing on their own in the market portfolio.

When its beta is 1.0, the project is no longer attractive because, as Figure 12.9 shows, its expected rate of return now lies below the security market line. The project offers a lower rate of return than investors can expect on equally risky investments. Therefore, it is negative-NPV. ■

⁹ See J. R. Graham and C. R. Harvey, "The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 (2001), pp. 187–243. A number of managers surveyed reported using more than one method to estimate the cost of capital. Seventy-three percent used the CAPM, while 39% stated they used the average historical stock return, and 34% used the CAPM with additional risk factors.

¹⁰ In Chapter 8, we described some special cases where you should prefer projects that offer a lower internal rate of return than the cost of capital. We assume here that your project is a "normal" one and that you prefer high IRRs to low ones.